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碩士論文

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Master’s Thesis

基於多重存取和多重資源的隨機存取通訊系統

Random Access Communications based on Multiple Access and Multiple Resources

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中文摘要

非正交的多重接取（NOMA）已被提議用來取代未來通信系統中的正交多重存取。一般而言，一些此領域知名的研究可以分為功率領域 NOMA、碼域 NOMA 和混合 NOMA。

分增益多重存取（GDMA）是將功率領域 NOMA 概念延伸至不僅僅考慮功率也考慮相位。在 GDMA 中，多個用戶可以共享單一資源，並且通過利用其不同的通道增益來區分接收到的信號中嵌入的傳送信號。

低密度簽名碼分碼多重存取（LDS-CDMA）及近期所提出的高密度簽名碼分碼多重存取（HDS-CDMA）可以歸類為碼域 NOMA。此外，這些多重存取技術還可以與低密度奇偶檢查（LDPC）碼和正交頻分多工（OFDM）結合使用。

隨機接取（Random Access）是無線通訊中的一部份，本文依據先前文獻所提出的透過分增益多重存取技術結合 ALOHA 的系統 (OFDM-GDMA-ALOHA) 和高密度簽名碼分碼多重存取技術結合 ALOHA 的系統 (OFDM-HDS-ALOHA) 進行結合並拓展。此外，我們考慮了一些重傳機制並對數據包的重傳次數設置了限制。

關鍵字：分增益多重接取、低密度簽記、高密度簽記、隨機存取

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Abstract

Gain Division Multiple Access (GDMA) extends the concept of power domain NOMA. Users simultaneously access the same resource and can be distinguished by leveraging their different channel gains.

Low-Density Signature Code Division Multiple Access (LDS-CDMA) and the recently proposed High-Density Signature Code Division Multiple Access (HDS-CDMA) are code domain NOMA. Additionally, these multiple access techniques can be combined with Low-Density Parity-Check (LDPC) codes and Orthogonal Frequency Division Multiplexing (OFDM).

Random Access is a part of wireless communications. This thesis focuses on the systems combining GDMA technology with ALOHA (OFDM-GDMA-ALOHA) and HDS-CDMA technology with ALOHA (OFDM-HDS-ALOHA). Moreover, we consider some retransmission mechanism and set a limit on the number of packet retransmissions. The situation with finite buffer is also studied.

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Keywords: GDMA, LDS-CDMA, HDS-CDMA, ALOHA system, Random access

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Chapter 1 Introduction

As technology rapidly advances and the multimedia era progresses, the demand for higher-speed and lower-latency network connections has surged, making high-speed wireless technology a significant research focus. Multiple Access (MA) techniques are crucial for boosting channel capacity and supporting more simultaneous connections in network systems. Traditionally, wireless networks have allocated radio resources to users using the orthogonal multiple access (OMA) principle, with examples like Code Division Multiple Access (CDMA) in third-generation (3G) systems and Orthogonal Frequency Division Multiple Access (OFDMA) [1] in fourth-generation (4G) systems. However, OMA systems face limitations as the number of users grows, leading to inefficiencies and reduced performance due to their reliance on orthogonal resources. To address these challenges, Non-Orthogonal Multiple Access (NOMA) [2] [3] has become an essential solution, as it removes the restriction on orthogonal subcarriers and enhances overall efficiency. In broad terms, notable research studies may be categorized into power-domain NOMA [3], code-domain NOMA [4] [5]. NOMA is widely implemented in advanced systems, such as PSMA [6] and PDMA [7], to better manage the increasing load and improve network performance.

In this thesis, we review GDMA [8], a gain-based NOMA system that distinguishes users by their individual channel gains. Unlike Power Domain NOMA (PD-NOMA),

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which superimposes users based on specific power allocations, GDMA utilizes variations in both amplitude and phase shifts caused by independent fading channels. The GDMA system model assumes that all U users transmit their messages simultaneously with perfect timing synchronization under independent fading channels. Consequently, there are 2^U possible superimposed signals at the receiver, and the probabilities can be calculated by analyzing the received signals and the superimposed signals.

LDS-CDMA [9] is a widely researched method within code domain NOMA. Its main advantage is the ability to support multiple simultaneous user transmissions within limited bandwidth over independent quasi-static fading channels. Typically, the message passing algorithm (MPA) is used with LDS-CDMA to iteratively decode transmitted data on the receiver side. Afterward, HDS-CDMA is proposed in [10]. The concept of HDS-CDMA involves each user transmitting messages to more than half of the total available resources. This approach enhances transmission efficiency, as it either requires fewer resources or achieves greater diversity, resulting in improved performance for a larger number of users across the entire system. Using HDS-CDMA, we can achieve greater diversity. Thus, HDS-CDMA has the benefits that it can require fewer resources without compromising error performance or provide better error performance with the same resource number. To reduce interference and improve spectral efficiency, We integrate all the above systems with the OFDM system [11] [12]. We also incorporate error-correcting codes (ECC) to achieve improved error performance.

Massive Machine Type Communication (mMTC) is designed to enable the connectivity of a vast number of users with sporadic transmission needs. The ALOHA system, a random access protocol, facilitates these sporadic transmissions. In a basic ALOHA system, collisions between multiple transmissions are treated as failures. Various ALOHA

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system variations have been proposed. We may integrate GDMA and HDS-CDMA into the ALOHA system [13] [12] to combat the collision problem. To maintain simplicity for devices in MTC, we exclusively utilize BPSK modulation. To reflect the practical access network, we further consider the retransmission of our slotted ALOHA systems and then consider our slotted ALOHA systems with finite buffer.

The structure of this thesis is as follows: Chapter 2 provides a review of GDMA concepts. In Chapter 3, both LDS-CDMA and HDS-CDMA are reviewed, along with the application of OFDM systems, error correction codes (ECC), and two distinct channel estimation algorithms to GDMA and HDS-CDMA. Chapter 4 focuses on various random access scheme. Finally, Chapter 5 presents the conclusions and outlines potential future research directions.

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Chapter 2 Gain Division Multiple Access

In this chapter, we consider the scenario for uplink transmissions over independent fading channels. We utilize GDMA [8], a gain-based NOMA system that distinguishes users by their individual channel gains. Overall, GDMA with multilevel detection and various constellation mappings are reviewed and shown.

2.1 Detection Principle

Assuming perfect timing synchronization, U users transmit signals to the same resource via independent fading channels. The block diagram is shown in Fig 2.1. The message bit sequence $\mathbf{c}_u$ is passed to the modulator with order m to produce the sequence of modulated symbols $\mathbf{x}_u = [x_u(1), x_u(2), \ldots, x_u(N_c/m)]$. The signals from all U users are transmitted to the receiver through independent Rayleigh flat-fading channels, and each received symbol in the sequence $\mathbf{r} = [r(1), r(2), \ldots, r(N_c/m)]$ can be described by equation (2.1)

$$r(i) = \sum_{u=1}^{U} h_u x_u(i) + w(i) = s(i) + w(i) \tag{2.1}$$

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Let h_u represent the complex channel coefficient of user u through the channel, $x_u(i)$ denote the i^{th} symbol sent from user u, , and $w(i)$ signify the complex additive white Gaussian noise (AWGN). The composite signal is given by $s(i)$. For simplicity, we omit the symbol index i in the rest of this thesis. Additionally, s can yield 2^{mU} possible levels.

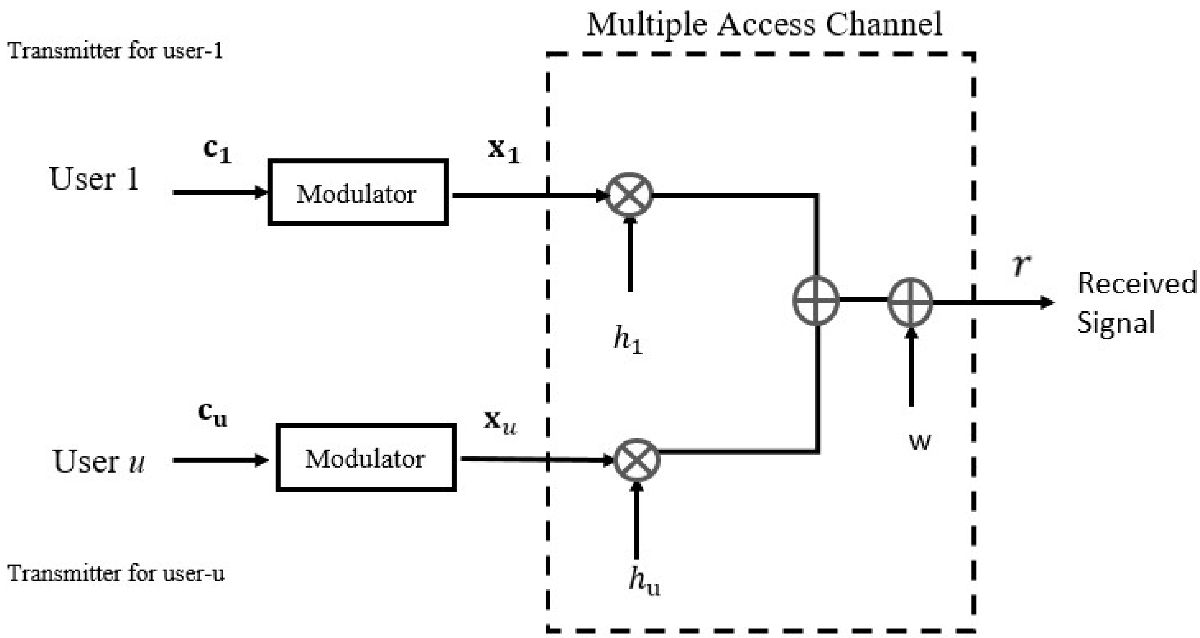

Figure 2.1: Transmitters of U users occupying the same resource.

The channel coefficient h_u for a Rician distributed channel can be expressed as:

$$h_u = \frac{1}{\sqrt{K+1}} \cdot a + \frac{\sqrt{K}}{\sqrt{K+1}} \cdot b \cdot i, \tag{2.2}$$

where a and b are normally distributed with a variance of 1 , with a representing the real part coefficient and b the imaginary part. When $K = 0$, h_u follows a Rayleigh distribution.

We break down the process of the GDMA detector into two steps. Initially, we calculate a 2^{mU} dimension probability vector of *a posteriori* probabilities (APPs) for all superimposed signals s. Subsequently, we derive the Log-Likelihood Ratio (LLR) for each user's bit based on these APP values. Accurate estimation of channel coefficients for all

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users is crucial to correctly determine the levels of superimposed signals. Once the Channel State Information (CSI) is perfectly recovered at the receiver, all possible levels of the received symbol s can be derived for further detection.

Take $U = 2$ and BPSK modulation ($m = 1$) as example, Figure 2.2 illustrates the block diagram of the GDMA detector, identifying users as user 1 and user 2. Table 2.1 outlines the mapping rules for the code bits of both users and the levels of superimposed signals denoted as $S[l]$, where $l \in \{0, 1, 2, 3\}$. Figure 2.3 is a graphical example of the superimposed levels.

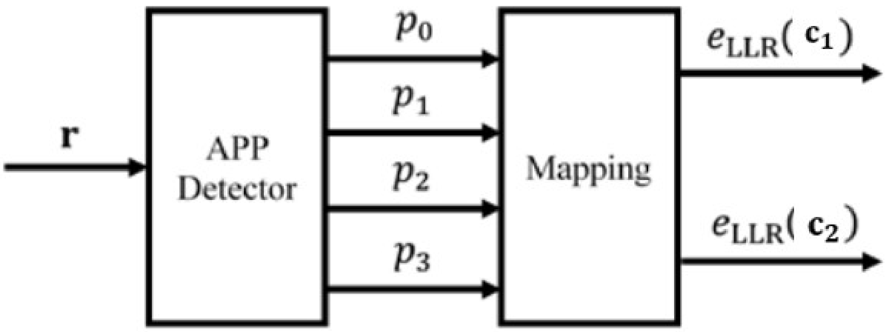

Figure 2.2: GDMA detector for BPSK and $U = 2$

l	X1	X2	h_1	h_2	S[l]
0	1	1	h_1	h_2	$h_1 + h_2$
1	1	-1	h_1	-h_2	$h_1 - h_2$
2	-1	1	-h_1	h_2	$-h_1 + h_2$
3	-1	-1	-h_1	-h_2	$-h_1 - h_2$

Table 2.1: Mapping rules for superimposed signal in the case of $m = 1$ and $U = 2$

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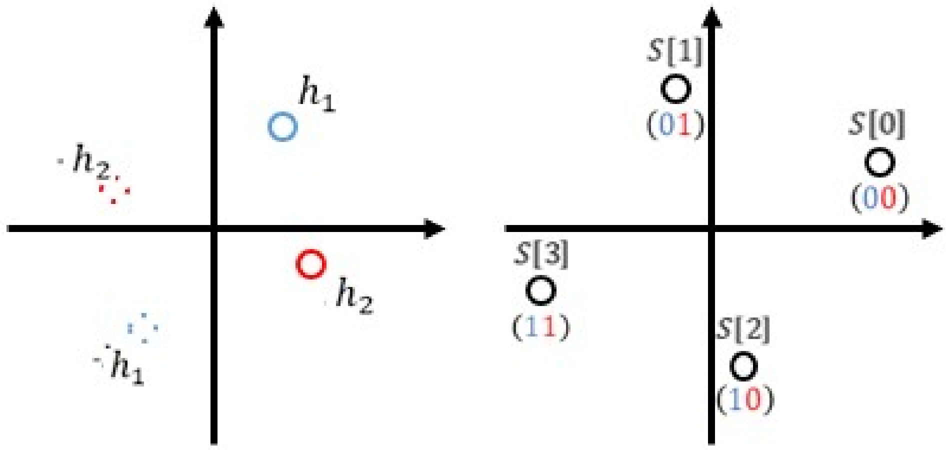

Figure 2.3: Superimposed levels in the case of BPSK and $U = 2$

The *a posteriori* probabilities (APPs) that s corresponds to the l^{th} level, with $l \in \{0, 1, 2, 3\}$, given the received symbol r, could be determined by

$$\begin{aligned} p_l &= \text{Pr}\{s = S[l]|r\} \\ &= \text{Pr}\{r|s = S[l]\} \cdot \frac{\text{Pr}\{s = S[l]\}}{\text{Pr}\{r\}}, \end{aligned} \tag{2.3}$$

while

$$\text{Pr}\{r|s = S[l]\} = \frac{1}{2\pi\sigma_w^2} \exp\left(-\frac{|r - S[l]|^2}{2\sigma_w^2}\right), \tag{2.4}$$

Let $\text{Pr}\{s = S[l]\}$ stands for the *a priori* probability of s and σ_w^2 is the variance of AWGN. According to the concept of physical layer network coding (PLNC), the APPs of code bits given r may be calculated as follows:

$$\begin{aligned} \text{Pr}\{c_1 = 0|r\} &= \text{Pr}\{s = S[0]|r\} + \text{Pr}\{s = S[1]|r\} = p_0 + p_1, \\ \text{Pr}\{c_1 = 1|r\} &= \text{Pr}\{s = S[2]|r\} + \text{Pr}\{s = S[3]|r\} = p_2 + p_3. \end{aligned} \tag{2.5}$$

and

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$$
\begin{aligned}
\Pr\{c_2 = 0|r\} &= \Pr\{s = S[0]|r\} + \Pr\{s = S[2]|r\} = p_0 + p_2, \\
\Pr\{c_2 = 1|r\} &= \Pr\{s = S[1]|r\} + \Pr\{s = S[3]|r\} = p_1 + p_3.
\end{aligned}
\tag{2.6}
$$

Once the APPs of code bits are acquired, the corresponding Log-Likelihood Ratios (LLRs) of code bits are determined by

$$
\begin{aligned}
e_{\mathrm{LLR}}(c_1) &= \ln \frac{\Pr\{c_1 = 0|r\}}{\Pr\{c_1 = 1|r\}} = \ln \frac{p_0 + p_1}{p_2 + p_3}, \\
e_{\mathrm{LLR}}(c_2) &= \ln \frac{\Pr\{c_2 = 0|r\}}{\Pr\{c_2 = 1|r\}} = \ln \frac{p_0 + p_2}{p_1 + p_3}.
\end{aligned}
\tag{2.7}
$$

The multi-level detection technique in a GDMA system can be readily extended to configurations with $U > 2$ users and $m > 1$ levels. For instance, in a scenario with $m = 1$ and $U = 3$, the mapping rules for the code bits of user 1 to user 3, along with the superimposed levels, are shown in Table 2.2. Each superimposed symbol s can have eight possible levels, as illustrated in Fig. 2.4. Let p_l for $l \in \{0, 1, \ldots, 7\}$ represent the *a posteriori* probabilities (APPs) of s given in Equation (2.3). The Log-Likelihood Ratios (LLRs) for the code bits are given by

$$
\begin{aligned}
e_{\mathrm{LLR}}(c_1) &= \ln \frac{\Pr\{c_1 = 0|r\}}{\Pr\{c_1 = 1|r\}} = \ln \frac{p_0 + p_1 + p_2 + p_3}{p_4 + p_5 + p_6 + p_7}, \\
e_{\mathrm{LLR}}(c_2) &= \ln \frac{\Pr\{c_2 = 0|r\}}{\Pr\{c_2 = 1|r\}} = \ln \frac{p_0 + p_1 + p_4 + p_5}{p_2 + p_3 + p_6 + p_7}, \\
e_{\mathrm{LLR}}(c_3) &= \ln \frac{\Pr\{c_3 = 0|r\}}{\Pr\{c_3 = 1|r\}} = \ln \frac{p_0 + p_2 + p_4 + p_6}{p_1 + p_3 + p_5 + p_7}.
\end{aligned}
\tag{2.8}
$$

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l	c_1	c_2	c_3	x_1	x_2	x_3	$S[l]$
0	0	0	0	1	1	1	$h_1 + h_2 + h_3$
1	0	0	1	1	1	-1	$h_1 + h_2 - h_3$
2	0	1	0	1	-1	1	$h_1 - h_2 + h_3$
3	0	1	1	1	-1	-1	$h_1 - h_2 - h_3$
4	1	0	0	-1	1	1	$-h_1 + h_2 + h_3$
5	1	0	1	-1	1	-1	$-h_1 + h_2 - h_3$
6	1	1	0	-1	-1	1	$-h_1 - h_2 + h_3$
7	1	1	1	-1	-1	-1	$-h_1 - h_2 - h_3$

Table 2.2: Mapping rules for superimposed signal in the case of $m = 1$ and $U = 3$

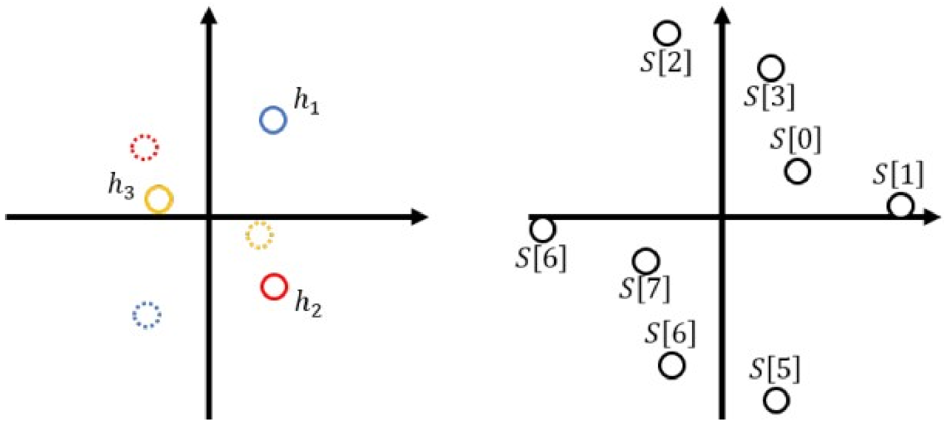

Figure 2.4: Superimposed levels in the case of BPSK and $U = 3$

2.2 Implementation Method

While the GDMA concept is straightforward, as the number of users and modulation methods increases, the number of superimposed signals also grows exponentially. In this section, we will elucidate systematic approaches for BPSK and QPSK modulation, for further modulation techniques, one can refer to [14].

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To broadly outline the application of GDMA MPSK, the detection process is delineated into three distinct phases: defining levels, deriving APP-values, and computing LLR-values.

2.2.1 GDMA BPSK implementation

In the first step, we convert the decimal number l_{10} into its binary counterpart l_2, where $l_2[u]$ denotes the u^{th} bit of the binary representation of l_{10}, where $u = 1, 2, \ldots, U$ and l_{10} ranges from 0 to $L-1$.

In detail:

$$\begin{aligned} 0_{10} &= (0, 0, \ldots, 0)_2, \\ 1_{10} &= (0, 0, \ldots, 1)_2, \\ &\vdots \\ (L-1)_{10} &= (1, 1, \ldots, 1)_2. \end{aligned} \tag{2.9}$$

The superimposed signals $S[l], l \in \{0, 1, \ldots, L-1\}$ can be expressed as

$$\begin{aligned} S[0] &= h_1 + h_2 + \ldots + h_u = (-1)^0 \cdot h_1 + (-1)^0 \cdot h_2 + \ldots + (-1)^0 \cdot h_u, \\ S[2] &= h_1 + h_2 + \ldots - h_u = (-1)^0 \cdot h_1 + (-1)^0 \cdot h_2 + \ldots + (-1)^1 \cdot h_u, \\ &\vdots \\ S[L-1] &= -h_1 - h_2 - \ldots - h_u = (-1)^1 \cdot h_1 + (-1)^1 \cdot h_2 + \ldots + (-1)^1 \cdot h_u. \end{aligned} \tag{2.10}$$

We can directly derive the mapping index from $S[l]$ by converting it from decimal to

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binary.

$$
\begin{aligned}
S\left[l_{10}\right] &= (-1)^{l_2[1]} \cdot h_1 + (-1)^{l_2[U]} \cdot h_2 + \cdots + (-1)^{l_2[U]} \cdot h_U \\
&= \sum_{u=1}^{U} (-1)^{l_2[u]} \cdot h_u, l_{10} = 0, 1, \cdots, L-1
\end{aligned}
\tag{2.11}
$$

where h_u represent the complex channel coefficient for user u.

In the second step, the *a posteriori* probabilities (APPs) for the l^{th} level, where $l \in \{0, 1, 2, \ldots, L-1\}$, can be computed as shown in Equation 2.3.

In the third step, the formula for calculating the Log-Likelihood Ratio (LLR) is provided as follows:

$$
\begin{aligned}
e_{\text{LLR}}\left[c_u\right] &= \ln \frac{\Pr\left\{c_u = 0 \mid r\right\}}{\Pr\left\{c_u = 1 \mid r\right\}} \\
&= \ln \frac{\sum_{l_{10}=0, l_2[u]\in 0}^{L} \Pr\{s = S[l] \mid r\}}{\sum_{l_{10}=0, l_2[u]\in 1}^{L} \Pr\{s = S[l] \mid r\}}
\end{aligned}
\tag{2.12}
$$

A hard decision can be made to determine whether a 0 or a 1 is transmitted by the user. The BER performance of uncoded GDMA-BPSK is illustrated in Figure 2.5.

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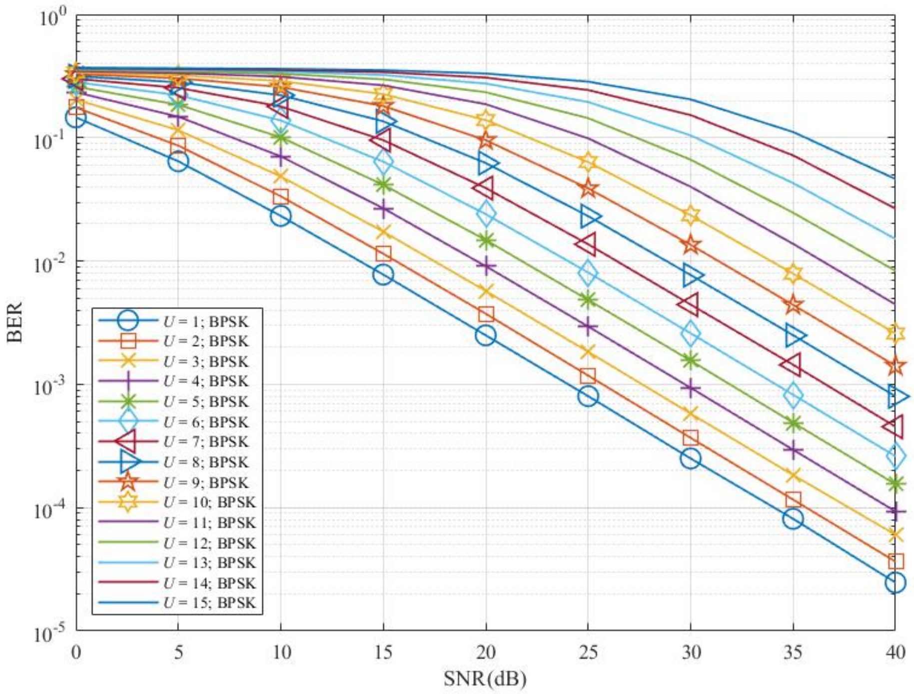

Figure 2.5: BER performances of uncoded GDMA-BPSK system over block Rayleigh fading channels.

2.2.2 GDMA QPSK implementation

In addition to BPSK, the detailed procedure for GDMA detection with QPSK (where $m = 2$) will also be discussed. Unlike BPSK, QPSK converts the decimal value l_{10} into a quaternary value l_4 due to its modulation order $m = 2$. In this context, $l_4[u]$ represents the u^{th} digit of the quaternary number l_4. Figure 2.6 illustrates the QPSK constellation based on Gray coding.

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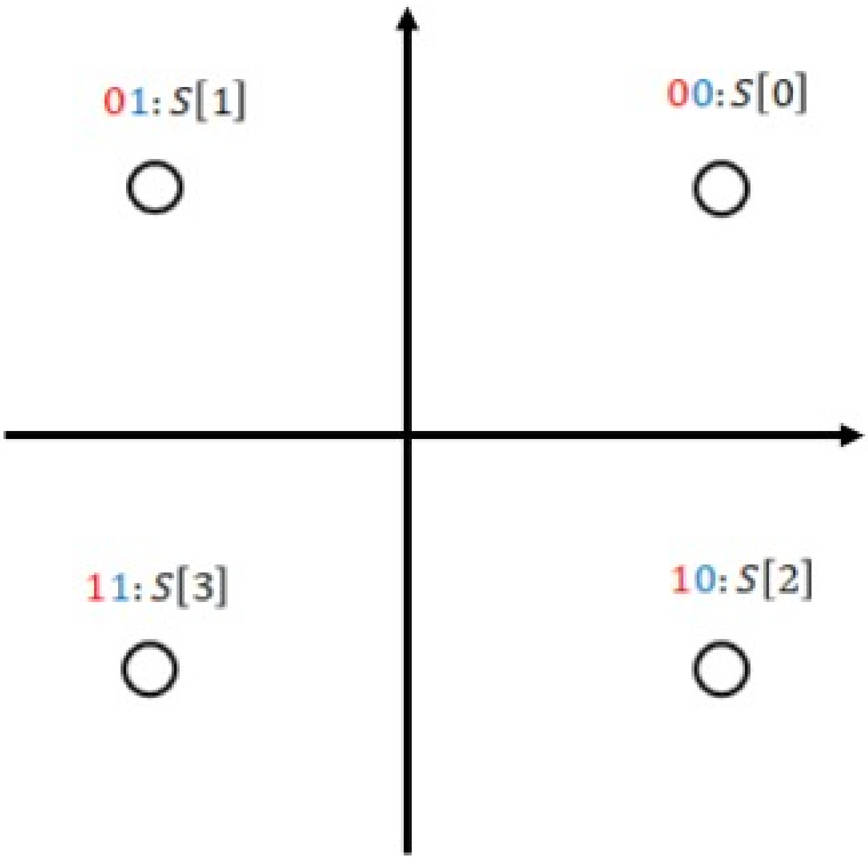

Figure 2.6: QPSK constellation based on Gray coding.

The superimposed signals $S[l], l \in \{0, 1, \ldots, L-1\}$ can be expressed as

$$\begin{aligned}
S[0] &= e^{\frac{\pi}{4}j}h_1 + e^{\frac{\pi}{4}j}h_2 + \cdots + e^{\frac{\pi}{4}j}h_U, \\
S[1] &= e^{\frac{\pi}{4}j}h_1 + e^{\frac{\pi}{4}j}h_2 + \cdots + e^{\left(\frac{\pi}{4}+\frac{\pi}{2}\right)j}h_U, \\
S[2] &= e^{\frac{\pi}{4}j}h_1 + e^{\frac{\pi}{4}j}h_2 + \cdots + e^{\left(\frac{\pi}{4}+\pi\right)j}h_U, \\
S[3] &= e^{\frac{\pi}{4}j}h_1 + e^{\frac{\pi}{4}j}h_2 + \cdots + e^{\left(\frac{\pi}{4}+\frac{3\pi}{2}\right)j}h_U, \\
&\vdots \\
S[L-1] &= e^{\left(\frac{\pi}{4}+\frac{3\pi}{2}\right)j}h_1 + e^{\left(\frac{\pi}{4}+\frac{3\pi}{2}\right)j}h_2 + \cdots + e^{\left(\frac{\pi}{4}+\frac{3\pi}{2}\right)j}h_U.
\end{aligned} \tag{2.13}$$

We can directly derive the mapping index from $S[l]$ by converting it from decimal to quaternary. Note that for the first bit ($i = 1$), both $S[0]$ and $S[1]$ map to a code bit of

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0, while $S[2]$ and $S[3]$ map to a code bit of 1. Here, i ranges from 1 to m, representing the i^{th} bit in the constellation. The mapping index $l_4[u]$ can be used to extract phase information from $S[l]$. Using this mapping principle, we can calculate LLR based on the i^{th} constellation position and user u.

$$
\begin{aligned}
S\left[l_{10}\right] &= e^{\left(\frac{\pi}{4}+\frac{\pi}{2}\cdot l_4[1]\right)j}\cdot h_1 + e^{\left(\frac{\pi}{4}+\frac{\pi}{2}\cdot l_4[1]\right)j}\cdot h_2 + \cdots + e^{\left(\frac{\pi}{4}+\frac{\pi}{2}\cdot l_4[1]\right)j}\cdot h_U \\
&= \sum_{u=1}^{U} e^{\left(\frac{\pi}{4}+\frac{\pi}{2}\cdot l_4[u]\right)j}\cdot h_u, l_{10}=0,1,\cdots,L-1; u=1,2,\cdots,U
\end{aligned}
\tag{2.14}
$$

Next, the *a posteriori* probabilities (APPs) for the l^{th} level, where $l \in \{0,1,2,\ldots,L-1\}$, can be computed as shown in Equation 2.3.

Last, the formula for calculating the Log-Likelihood Ratio (LLR) is provided as follows:

$$
e_{\text{LLR}}\left[c_u\right] = \ln\frac{\Pr\left\{c_u=0\mid r\right\}}{\Pr\left\{c_u=1\mid r\right\}} = \begin{cases} \ln\frac{\sum_{l_{10}=0,l_2[u]\in\{0,1\}}^{L}\Pr\{s=S[l]|r\}}{\sum_{l_{10}=0,l_2[u]\in\{2,3\}}^{L}\Pr\{s=S[l]|r\}}, i=1; u=1,\cdots,U \\ \ln\frac{\sum_{l_{10}=0,l_2[u]\in\{0,2\}}^{L}\Pr\{s=S[l]|r\}}{\sum_{l_{10}=0,l_2[u]\in\{1,3\}}^{L}\Pr\{s=S[l]|r\}}, i=2; u=1,\cdots,U \end{cases}
\tag{2.15}
$$

The BER performance of uncoded GDMA-QPSK is illustrated in Figure 2.7.

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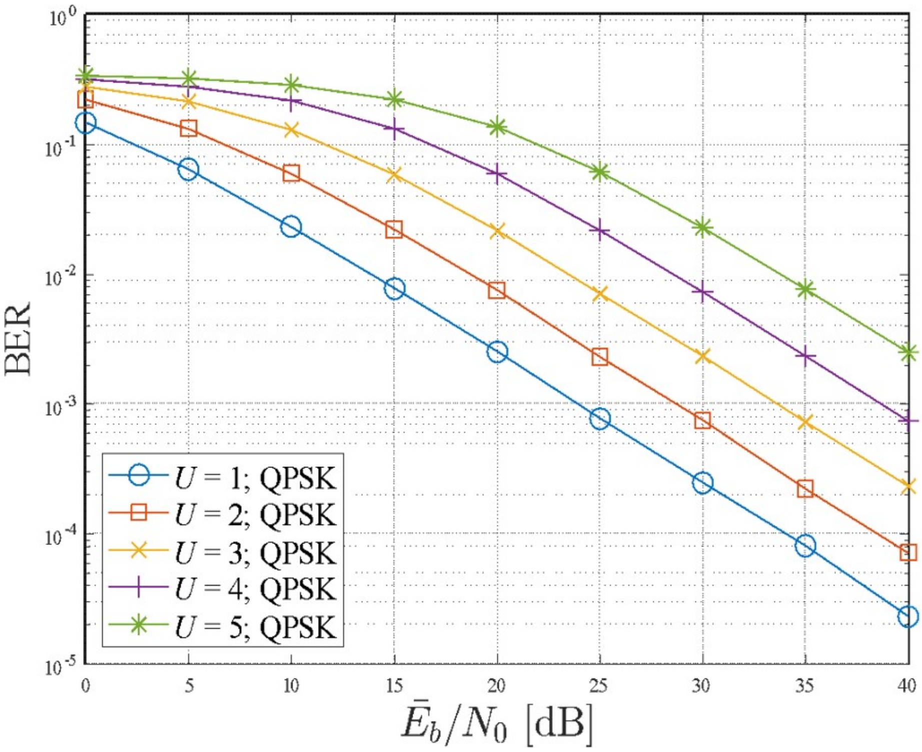

Figure 2.7: BER performances of uncoded GDMA-QPSK system over block Rayleigh fading channels.

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Chapter 3 LDS-CDMA and HDS-CDMA

The LDS-CDMA scheme can be viewed as an amalgamation of multiple dependent GDMA systems, each applied to a different resource. The initial message passing for LDS-CDMA can similarly be understood as the communication between various GDMA systems to utilize their interdependencies. Building on this interpretation, [10] develop high-density (HDS) CDMA schemes. The benefit of HDS-CDMA schemes lies in their ability to either require fewer resources or achieve improved error performance. The above concepts may be integrated into OFDM.

3.1 LDS-CDMA

In traditional code division multiple access, U users operate within the same time slot or frequency band assignments, with every user utilizing a distinct spreading sequence. This configuration implies in a traditional CDMA system, N resources are used, where $U = N$. Low density signature CDMA (LDS-CDMA) and high density signature CDMA (HDS-CDMA), as described in [9] and [10], are categorized as code domain NOMA. In both cases, a user selects some, but not all, of the available resources, so each resource is

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only partially occupied by the users. Thus, U may exceed N, with each resource being occupied by $\hat{U}$ users. In this thesis, we categorize the system as LDS-CDMA when the ratio $\frac{\hat{U}}{U}$ is less than or equal to $\frac{1}{2}$, and as HDS-CDMA when the ratio $\frac{\hat{U}}{U}$ exceeds $\frac{1}{2}$. We introduce a unified representation and may denote both cases as GS (Generalized Signature) in our simulation graphs. The representation comprises numbers of users of each diversities since we found out their performance heavily depends on diversities in previous work. Take a conventional GDMA system with U users as example, we denote it as a $(1, U, U)$ system. The first parameter acts for the resource number, the second number represents the user number, and the last parameter speaks for U users of diversity 1. In Figure 3.1, it is a (4,6,0,6,0,0) LDS-CDMA system, the representation points out that there are 4 resources, 6 users, 0 user of diversity 1, 6 users of diversity 2, 0 user of diversity 3, and 0 user of diversity 4 in the system. There is extensive research on LDS-CDMA. In [15], the author estimates BER performance of a LDS-CDMA system by a GDMA system. In [13], the LDS-CDMA system is conceptualized as an amalgamation of multiple interconnected GDMA systems. Given the interdependency of GDMA systems in LDS-CDMA, MPA is commonly employed for decoding by transmitting information derived from these interconnected GDMA systems.

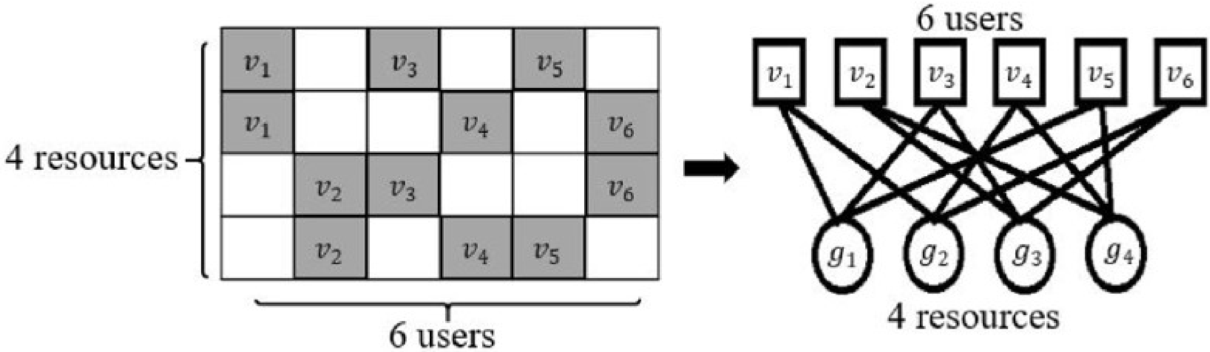

Figure 3.1: Encoder of LDS-CDMA along with its associated Tanner Graph

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3.1.1 System Model and Encoding

Consider the $(4, 6, 0, 6, 0, 0)$ LDS-CDMA system as stated before illustrated in Fig. 3.1, which also includes the corresponding Tanner graph. In the left graph, the rows represent the 4 distinct resources and the columns represent the 6 different users. Each gray block indicates a resource occupied by a user. This block graph can be transformed into a Tanner graph by linking the connections between the resources and the users. It can be observed that each resource includes three users, and each user is associated with two resources. Consequently, each user experiences a diversity of 2 during detection.

3.1.2 Decoding Process

A message passing algorithm may be employed to decode the LDS-CDMA message due to the Tanner graph structure. The decoding process involves three steps, as outlined below. To get a more complete understanding of the entire process, we will also use the $(4, 6, 0, 6, 0, 0)$-BPSK case as an example.

3.1.2.1 Initialization

Assume all transmitted bits are equally likely to be 0 or 1. Then, the a priori probability of each message combination is $\frac{1}{M}$, where M is the total number of possible message combinations for a user. Thus, the a priori probability of each message for the u-th user in the n-th resource is $P_{u_u \to r_n}(m_u) = \text{apr}(m_u) = \frac{1}{M}$, where $M = 2^m$ and m is the modulation level. Take the first user to the first resource of $(4, 6, 0, 6, 0, 0)$-BPSK as example, we initialize $P_{u_1 \to r_A}(u_1 = 0) = \frac{1}{2}$ and $P_{u_1 \to r_A}(u_1 = 1) = \frac{1}{2}$.

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3.1.2.2 Iteration

The iteration phase is divided into two components: resource-node update and user-node update. In the Tanner graph, a resource node r_n is connected to three user nodes u_u. The received signal at resource r_n is given by

$$y_n = \sum_{u=1}^{6} h_{u,n} x_u(m_u) + w_n,$$

where $h_{u,n}$ denotes the channel fading between r_n and u_u. If there is no connection between them, $h_{u,n}$ is zero. The probability $P(y_n|\mathbf{m})$ can be similarly determined using multi-level detection. Take resource r_n from Fig. 3.1 as an example. For simplicity, the resource-node update formula reindexes the connected user nodes from 1 to 3 as follows:

$$P_{r_n \to u_1}(m_1) = \sum_{m_2=0}^{M-1} \sum_{m_3=0}^{M-1} P(y_n|\mathbf{m})(P_{u_2 \to r_n}(m_2) P_{u_3 \to r_n}(m_3)) \tag{3.1}$$

$$P_{r_n \to u_2}(m_2) = \sum_{m_1=0}^{M-1} \sum_{m_3=0}^{M-1} P(y_n|\mathbf{m})(P_{u_1 \to r_n}(m_1) P_{u_3 \to r_n}(m_3)) \tag{3.2}$$

$$P_{r_n \to u_3}(m_3) = \sum_{m_1=0}^{M-1} \sum_{m_2=0}^{M-1} P(y_n|\mathbf{m})(P_{u_1 \to r_n}(m_1) P_{u_2 \to r_n}(m_2)) \tag{3.3}$$

Equations (3.1), (3.2), and (3.3) apply to a system with 6 users, 4 resources. Here, $\mathbf{m} = (m_1, m_2, m_3)$ represents the possible messages for users, and $P_{r_n \to u_u}$ denotes the information from the resource node to user u. For instance, in equation (3.1), $P_{r_n \to u_1}$ contains the probabilities of the possible messages for user 1, while $P_{u_2 \to r_n}$ and $P_{u_3 \to r_n}$ provide extrinsic information from user 2 and user 3, respectively.

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Subsequently, each user node passes updates obtained from extrinsic information back to its neighboring resource nodes using the following equations. The normalization function is used to ensure that the total probability for a user is 1.

$$P_{u1 \to r_A}(m_u) = \text{Normalize}(P_{r_B \to u_1}(m_u)), \tag{3.4}$$

$$P_{u1 \to r_B}(m_u) = \text{Normalize}(P_{r_A \to u_1}(m_u)). \tag{3.5}$$

The algorithm proceeds by alternately updating resource nodes and user nodes until it either converges to equilibrium or the maximum iteration limit is reached.

3.1.2.3 Decision

Finally, we calculate the LLR-values for each user node using the following equations. Here, $Q_v(m_u)$ represents the probability of message m for user node u after multiplying all edge values and the a priori probability.

$$\text{LLR}_{u,b_i} = \ln\left(\frac{Q_v(m_u)|_{b_i=1}}{Q_v(m_u)|_{b_i=0}}\right), \tag{3.6}$$

where

$$Q_v(m_u) = (P_{r_A \to u_u}(m_u))(P_{r_B \to u_u}(m_u)). \tag{3.7}$$

for simplicity, we reindex the connected resource nodes to r_A and r_B and take Fig. 3.1 as

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an example.

3.2 HDS-CDMA

The thesis [10] introduced High-Density Signature Code Division Multiple Access (HDS-CDMA), which integrates the principles of GDMA and LDS-CDMA. In HDS-CDMA, each user transmits messages to more than half of the available resources, aiming to enhance system diversity and improve error performance in fading channels. In this section, we will explore various transmission arrangement systems and different decoding algorithms. Finally, we will compare the error rate performance across these systems.

3.2.1 (2,3,0,3) HDS-CDMA and (4,6,0,0,0,6) HDS-CDMA

We will review two renowned HDS-CDMA systems, (2,3,0,3) HDS-CDMA and (4,6,0,0,0,6) HDS-CDMA, one may refer to [13] for further complicated cases. As discussed in the previous section, the first parameter represents the total number of resources, the second parameter denotes the total number of users, and the rest parameters indicate the number of users of each diversity. Therefore, in a (2,3,0,3) HDS-CDMA system, there are 2 resources and 3 users, with all 3 users transmitting in each resource. Similarly, in a (4,6,0,0,0,6) HDS-CDMA system, there are 6 users and 4 resources, with all 6 users transmitting in each resource. In Fig. 3.2, the encoder along with the Tanner graph for the (2,3,0,3) HDS-CDMA system are depicted, while those for the (4,6,0,0,0,6) HDS-CDMA system are illustrated in Fig. 3.3.

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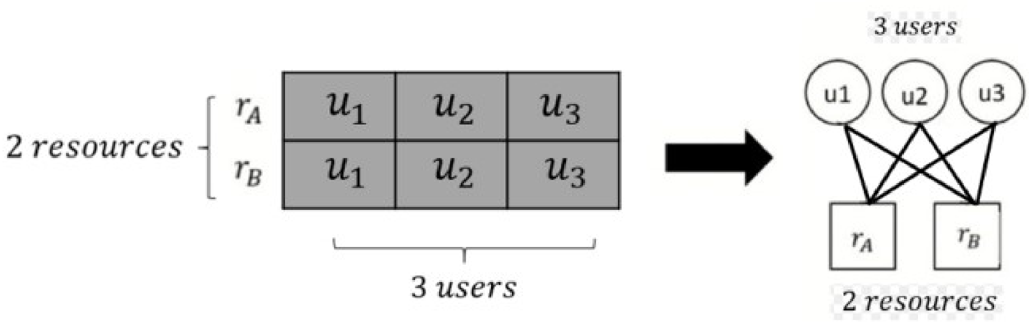

Figure 3.2: A (2,3,0,3) encoder along with its associated Tanner Graph

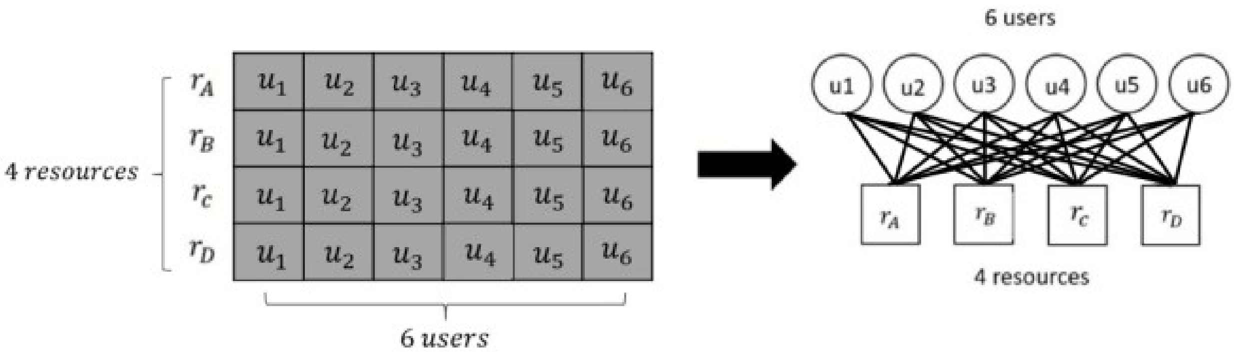

Figure 3.3: A (4,6,0,0,0,6) encoder along with its associated Tanner Graph

3.2.2 Decoding Algorithms

We will employ several different algorithms for decoding and compare their error performance.

3.2.2.1 First Algorithm : IND Decoding

The (2,3,0,3) HDS-CDMA system can be considered as two 3-user GDMA systems. After performing GDMA detection on resource A and resource B, we obtain the LLR-values for all bits of all users on r_A and r_B. Since the two resources experience different fading channels, the "IND decoding" approach involves adding the LLR-values calculated from both resources to produce our detection results.

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Similarly, the (4,6,0,0,0,6) HDS-CDMA system can be viewed as comprising four 6-user GDMA systems. By conducting GDMA detection on resource A, resource B, resource C, and resource D, we acquire the LLR-values for all bits of every user on r_A, r_B, r_C, and r_D. Given that the four resources are subject to different fading channels, the "IND decoding" method entails summing the LLR-values obtained from all resources to generate our detection outcomes.

This method helps mitigate the channel gain instability caused by fading, and the visualization of the (2,3,0,3) and (4,6,0,0,0,6) decoding concept can easily be seen in Fig. 3.4 and Fig. 3.5 respectively.

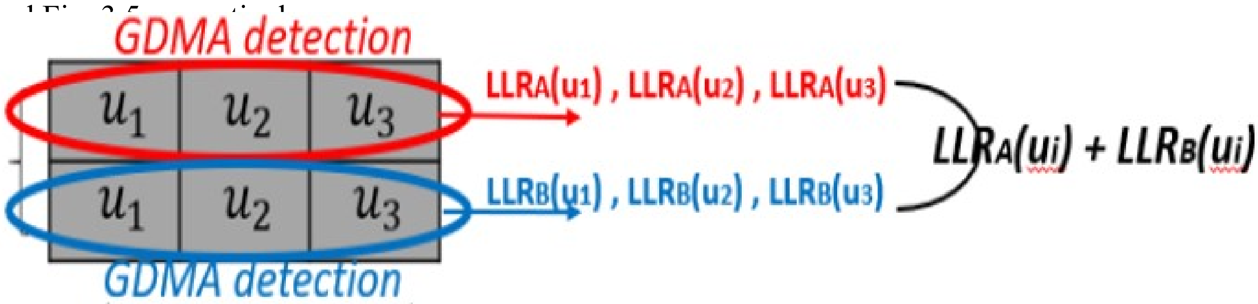

Figure 3.4: Visualization of IND decoding of (2,3,0,3) HDS-CDMA

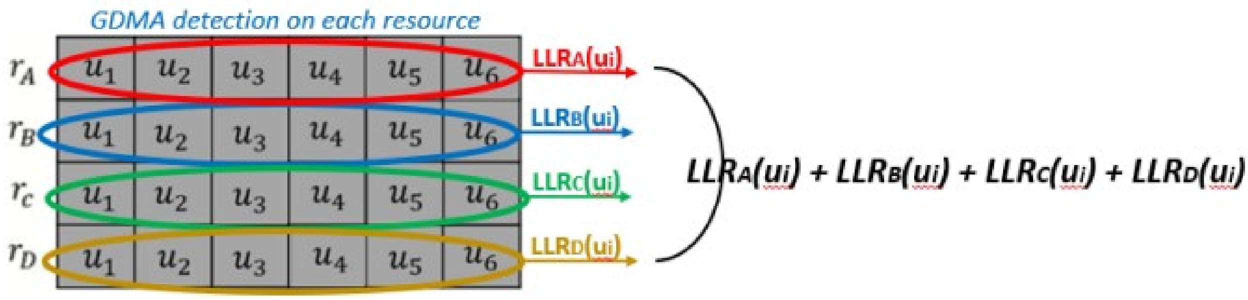

Figure 3.5: Visualization of IND decoding of (4,6,0,0,0,6) HDS-CDMA

3.2.2.2 Second Algorithm : MAP and ML Decoding

Due to the suboptimal BER performance observed with the IND algorithm, we opted to utilize the MAP algorithm for decoding. Prior to introducing the MAP algorithm, it is essential to discuss the Bayesian decision theorem. This theorem involves estimating

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unknown states based on subjective probabilities in situations of incomplete information. The Bayesian formula is then employed to adjust these probabilities, ultimately guiding optimal decision-making through expected values and refined probabilities.

Basic idea of Bayesian Decision Theory:

The basic idea of the Bayesian decision theorem, particularly in the context of Maximum A Posteriori (MAP) detection, involves making decisions that minimize the expected risk based on the posterior probabilities of the hypotheses. Here is a concise breakdown of how this works:

1. **Bayesian Framework**:

   - In the Bayesian approach, we incorporate prior knowledge about the hypotheses along with the likelihood of the observed data given these hypotheses.

   - We compute the posterior probability $P(H_i \mid D)$ of each hypothesis H_i given the data D.

2. **Posterior Probability**:

   - The posterior probability is given by Bayes' theorem:

$$P(H_i \mid D) = \frac{P(D \mid H_i)P(H_i)}{P(D)}$$

   - Here, $P(D \mid H_i)$ is the likelihood, $P(H_i)$ is the prior probability, and $P(D)$ is the marginal likelihood or evidence.

3. **Decision Rule**:

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- The MAP decision rule selects the hypothesis that maximizes the posterior probability:

$$\hat{H}_{\text{MAP}} = \arg\max_{H_i} P(H_i \mid D)$$

- This means we choose the hypothesis H_i for which $P(H_i \mid D)$ is highest.

4. **Expected Risk**:

- The expected risk (or expected loss) is minimized by choosing the hypothesis with the highest posterior probability, assuming that the loss function is 0-1 (i.e., the loss is 1 if we make an incorrect decision and 0 if we make a correct decision).

Summary

The Bayesian decision theorem in the context of MAP estimation focuses on utilizing prior information and the observed data to compute posterior probabilities for different hypotheses. The decision is then made by selecting the hypothesis with the highest posterior probability, thereby minimizing the expected risk under the assumption of a simple loss function.

In our decoding process, ”MAP Decoding Algorithm” is divided into two main steps. First, we calculate *a posterior* probabilities. Second, LLR-values can be calculated by *a posterior* probabilities. For simplicity, we will take the first user of these systems as an example.

First step:

The equation of calculating app_{u_1} of (2,3,0,3) HDS-CDMA is shown in equation (3.8).

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$$app_{u_1}(u_1=0)=p(u_1=0\mid r_A,r_B)=\frac{p(u_1=0)p(r_A,r_B\mid u_1=0)}{p(r_A,r_B)},$$

$$app_{u_1}(u_1=1)=p(u_1=1\mid r_A,r_B)=\frac{p(u_1=1)p(r_A,r_B\mid u_1=1)}{p(r_A,r_B)}. \tag{3.8}$$

The equation of calculating app_{u_1} of (4,6,0,0,0,6) HDS-CDMA is shown in equation (3.9).

$$app_{u_1}(u_1=0)=p(u_1=0\mid r_A,r_B,r_C,r_D)=\frac{p(u_1=0)p(r_A,r_B,r_C,r_D\mid u_1=0)}{p(r_A,r_B,r_C,r_D)},$$

$$app_{u_1}(u_1=1)=p(u_1=1\mid r_A,r_B,r_C,r_D)=\frac{p(u_1=1)p(r_A,r_B,r_C,r_D\mid u_1=1)}{p(r_A,r_B,r_C,r_D)}. \tag{3.9}$$

where $p(u_1=0)=p(u_1=1)=\frac{1}{M}, M=2^m$, m is modulation level, $p(r_A,r_B\mid u_1)$ and $p(r_A,r_B,r_C,r_D\mid u_1)$ are likelihood functions, $p(r_A,r_B)$ and $p(r_A,r_B,r_C,r_D)$ are full probabilities.

Second step:

The equation of calculating LLR_{u_1} of $(2,3,0,3)$ HDS-CDMA is shown in equation (3.10).

$$\begin{aligned} LLR_{u_1} &= \ln\left(\frac{p(u_1=0|r_A,r_B)}{p(u_1=1|r_A,r_B)}\right) \\ &= \ln\left(\frac{\sum_{u_2}\sum_{u_3}p(u_1=0,u_2,u_3|r_A,r_B)}{\sum_{u_2}\sum_{u_3}p(u_1=1,u_2,u_3|r_A,r_B)}\right) \end{aligned} \tag{3.10}$$

The equation of calculating LLR_{u_1} of $(4,6,0,0,0,6)$ HDS-CDMA is shown in equation (3.11).

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$$
\begin{aligned}
LLR_{u_1} &= \ln\left(\frac{p(u_1 = 0|r_A, r_B, r_C, r_D)}{p(u_1 = 1|r_A, r_B, r_C, r_D)}\right) \\
&= \ln\left(\frac{\sum_{u_2,u_3,u_4,u_5,u_6} p(u_1 = 0, u_2, u_3, u_4, u_5, u_6|r_A, r_B, r_C, r_D)}{\sum_{u_2,u_3,u_4,u_5,u_6} p(u_1 = 1, u_2, u_3, u_4, u_5, u_6|r_A, r_B, r_C, r_D)}\right)
\end{aligned} \tag{3.11}
$$

Transition to ML Decoder

For simplicity, in our simulation, we suppose *a prior* probability of all transmitted bits are equally likely to be 0 or 1. Under the assumption of equal prior probabilities, the Bayesian decision rule simplifies to the Maximum Likelihood (ML) decoder. Since $P(H_i)$ is the same for all hypotheses, the decision rule depends only on the likelihood $P(D \mid H_i)$:

- The ML decoder selects the hypothesis that maximizes the likelihood:

$$
\hat{H}_{\text{ML}} = \arg\max_{H_i} P(D \mid H_i)
$$

- This means we choose the hypothesis H_i for which $P(D \mid H_i)$ is highest, ignoring the prior probabilities as they are equal.

The Bayesian decision theorem in the context of MAP estimation focuses on utilizing prior information and the observed data to compute posterior probabilities for different hypotheses. Assuming equal prior probabilities, the MAP decision rule simplifies to the ML decoder, which selects the hypothesis that maximizes the likelihood of the observed data.

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Therefore, for ML decoder, we can simplify equation (3.10) to equation (3.12).

$$
\begin{aligned}
LLR_{u_1} &= \ln\left(\frac{p(r_A|u_1=0)p(r_B|u_1=0)}{p(r_A|u_1=1)p(r_B|u_1=1)}\right) \\
&= \ln\left(\frac{\sum_{u_2}\sum_{u_3} p(r_A|u_1=0,u_2,u_3)p(r_B|u_1=0,u_2,u_3)}{\sum_{u_2}\sum_{u_3} p(r_A|u_1=1,u_2,u_3)p(r_B|u_1=1,u_2,u_3)}\right)
\end{aligned}
\tag{3.12}
$$

3.2.2.3 Third Algorithm : MPA Decoding

Similar to LDS-CDMA, message passing algorithm can also be employed to decode HDS-CDMA. We will briefly state the process.

Initialization

Assume all transmitted bits are equally likely to be 0 or 1. Then, the a priori probability of each message combination is $\frac{1}{M}$, where M is the total number of possible message combinations for a user. Thus, the a priori probability of each message for the u-th user in the n-th resource is $P_{u_u \to r_n}(m_u) = \text{apr}(m_u) = \frac{1}{M}$, where $M = 2^m$ and m is the modulation level.

Iteration

The iteration phase is divided into two components: resource-node update and user-node update. For HDS-CDMA in our example, the resource node r_n connects all user nodes u_u. We also have the received signal $y_n = \sum_{u=1}^{N} h_{u,n} x_u(m_u) + w_n$ at resource r_n. Note that $h_{u,n} = 0$ if there is no connection between them. However, in our example, $h_{u,n} = 1$ for all conditions. One can refer to [10] for more complicated cases. The probability $P(y_n|\mathbf{m})$ can be similarly determined using multi-level detection. Take resource r_n

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from Fig. 3.2 and Fig. 3.3 as an example.

For resource r_n from (2,3,0,3) system from Fig. 3.2

$$P_{r_n \to u_1}(m_1) = \sum_{m_2=0}^{M-1} \sum_{m_3=0}^{M-1} P(y_n|\mathbf{m})(P_{u_2 \to r_n}(m_2) P_{u_3 \to r_n}(m_3)) \tag{3.13}$$

$$P_{r_n \to u_2}(m_2) = \sum_{m_1=0}^{M-1} \sum_{m_3=0}^{M-1} P(y_n|\mathbf{m})(P_{u_1 \to r_n}(m_1) P_{u_3 \to r_n}(m_3)) \tag{3.14}$$

$$P_{r_n \to u_3}(m_3) = \sum_{m_1=0}^{M-1} \sum_{m_2=0}^{M-1} P(y_n|\mathbf{m})(P_{u_1 \to r_n}(m_1) P_{u_2 \to r_n}(m_2)) \tag{3.15}$$

For resource r_n from (4,6,0,0,0,6) system from Fig. 3.3

$$P_{r_n \to u_1}(m_1) = \sum_{m_2=0}^{M-1} \sum_{m_3=0}^{M-1} \sum_{m_4=0}^{M-1} \sum_{m_5=0}^{M-1} \sum_{m_6=0}^{M-1} P(y_n|\mathbf{m}) \\ (P_{u_2 \to r_n}(m_2) P_{u_3 \to r_n}(m_3) P_{u_4 \to r_n}(m_4) P_{u_5 \to r_n}(m_5) P_{u_6 \to r_n}(m_6)) \tag{3.16}$$

$$P_{r_n \to u_2}(m_2) = \sum_{m_1=0}^{M-1} \sum_{m_3=0}^{M-1} \sum_{m_4=0}^{M-1} \sum_{m_5=0}^{M-1} \sum_{m_6=0}^{M-1} P(y_n|\mathbf{m}) \\ (P_{u_1 \to r_n}(m_1) P_{u_3 \to r_n}(m_3) P_{u_4 \to r_n}(m_4) P_{u_5 \to r_n}(m_5) P_{u_6 \to r_n}(m_6)) \tag{3.17}$$

$$P_{r_n \to u_3}(m_3) = \sum_{m_1=0}^{M-1} \sum_{m_2=0}^{M-1} \sum_{m_4=0}^{M-1} \sum_{m_5=0}^{M-1} \sum_{m_6=0}^{M-1} P(y_n|\mathbf{m}) \\ (P_{u_1 \to r_n}(m_1) P_{u_2 \to r_n}(m_2) P_{u_4 \to r_n}(m_4) P_{u_5 \to r_n}(m_5) P_{u_6 \to r_n}(m_6)) \tag{3.18}$$

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$$P_{r_n \to u_4}(m_4) = \sum_{m_1=0}^{M-1} \sum_{m_2=0}^{M-1} \sum_{m_3=0}^{M-1} \sum_{m_5=0}^{M-1} \sum_{m_6=0}^{M-1} P(y_n|\mathbf{m})$$

$$(P_{u_1 \to r_n}(m_1) P_{u_2 \to r_n}(m_2) P_{u_3 \to r_n}(m_3) P_{u_5 \to r_n}(m_5) P_{u_6 \to r_n}(m_6)) \quad (3.19)$$

$$P_{r_n \to u_5}(m_5) = \sum_{m_1=0}^{M-1} \sum_{m_2=0}^{M-1} \sum_{m_3=0}^{M-1} \sum_{m_4=0}^{M-1} \sum_{m_6=0}^{M-1} P(y_n|\mathbf{m})$$

$$(P_{u_1 \to r_n}(m_1) P_{u_2 \to r_n}(m_2) P_{u_3 \to r_n}(m_3) P_{u_4 \to r_n}(m_4) P_{u_6 \to r_n}(m_6)) \quad (3.20)$$

$$P_{r_n \to u_6}(m_6) = \sum_{m_1=0}^{M-1} \sum_{m_2=0}^{M-1} \sum_{m_3=0}^{M-1} \sum_{m_4=0}^{M-1} \sum_{m_5=0}^{M-1} P(y_n|\mathbf{m})$$

$$(P_{u_1 \to r_n}(m_1) P_{u_2 \to r_n}(m_2) P_{u_3 \to r_n}(m_3) P_{u_4 \to r_n}(m_4) P_{u_5 \to r_n}(m_5)) \quad (3.21)$$

Equations (3.13) to (3.15) apply to a system with 3 users, 2 resources, and (3.16) to (3.21) apply to a system with 6 users, 4 resources. Here, **m** represents the possible messages for users, and $P_{r_n \to u_u}$ denotes the information from the resource node to user u. For instance, in equation (3.13), $P_{r_n \to u_1}$ contains the probabilities of the possible messages for user 1, while $P_{u_2 \to r_n}$ and $P_{u_3 \to r_n}$ provide extrinsic information from user 2 and user 3, respectively.

Subsequently, each user node passes updates obtained from extrinsic information back to its neighboring resource nodes using the following equations. The normalization function is used to ensure that the total probability for a user is 1.

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To update $P_{u_1 \to r_n}(m_u)$ of (2,3,0,3) system, the equations are as follows.

$$P_{u_1 \to r_A}(m_u) = \text{Normalize}(P_{r_B \to u_1}(m_u)), \tag{3.22}$$

$$P_{u_1 \to r_B}(m_u) = \text{Normalize}(P_{r_A \to u_1}(m_u)). \tag{3.23}$$

Similarly, to update $P_{u_1 \to r_n}(m_u)$ of (4,6,0,0,0,6) system, the equations are as follows.

$$P_{u_1 \to r_A}(m_u) = \text{Normalize}(P_{r_B \to u_1}(m_u)P_{r_C \to u_1}(m_u)P_{r_D \to u_1}(m_u)), \tag{3.24}$$

$$P_{u_1 \to r_B}(m_u) = \text{Normalize}(P_{r_A \to u_1}(m_u)P_{r_C \to u_1}(m_u)P_{r_D \to u_1}(m_u)), \tag{3.25}$$

$$P_{u_1 \to r_C}(m_u) = \text{Normalize}(P_{r_A \to u_1}(m_u)P_{r_B \to u_1}(m_u)P_{r_D \to u_1}(m_u)), \tag{3.26}$$

$$P_{u_1 \to r_D}(m_u) = \text{Normalize}(P_{r_A \to u_1}(m_u)P_{r_B \to u_1}(m_u)P_{r_C \to u_1}(m_u)). \tag{3.27}$$

The algorithm proceeds by alternately updating resource nodes and user nodes until it either converges to equilibrium or the maximum iteration limit is reached.

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Decision

Finally, the LLR-values for each user node can be obtained using the following equations. Here, $Q_v(m_u)$ represents the probability of message m for user node u after multiplying all edge values and the a priori probability.

$$\mathrm{LLR}_{u,b_i} = \ln\left(\frac{Q_v(m_u)|_{b_i=1}}{Q_v(m_u)|_{b_i=0}}\right), \tag{3.28}$$

where

$$Q_v(m_u) = (P_{r_A \to u_u}(m_u))(P_{r_B \to u_u}(m_u)), \tag{3.29}$$

and

$$Q_v(m_u) = (P_{r_A \to u_u}(m_u))(P_{r_B \to u_u}(m_u))(P_{r_C \to u_u}(m_u))(P_{r_D \to u_u}(m_u)). \tag{3.30}$$

for (2,3,0,3) HDS-CDMA system and (4,6,0,0,0,6) HDS-CDMA system respectively.

3.3 Simulation Results of LDS-CDMA and HDS-CDMA

In order to facilitate subsequent comparisons, we briefly review transmission efficiency. We define the transmission efficiency by $\frac{Um}{N}$ bits/resource. We will list some examples. The $(1, 3, 3)$ GDMA system achieves a transmission efficiency of 3 bits per resource unit when utilizing BPSK , and this efficiency increases to 6 bits per resource

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unit with QPSK modulation. On the other hand, the $(4,6,0,6,0,0)$ LDS-CDMA system delivers a transmission efficiency of 1.5 bits per resource unit with BPSK, this efficiency increases to 3 bits per resource unit with QPSK modulation. We assume all U users experience independent Rayleigh fading for all N resources. We also assume perfect CSI. The BER performances are shown below.

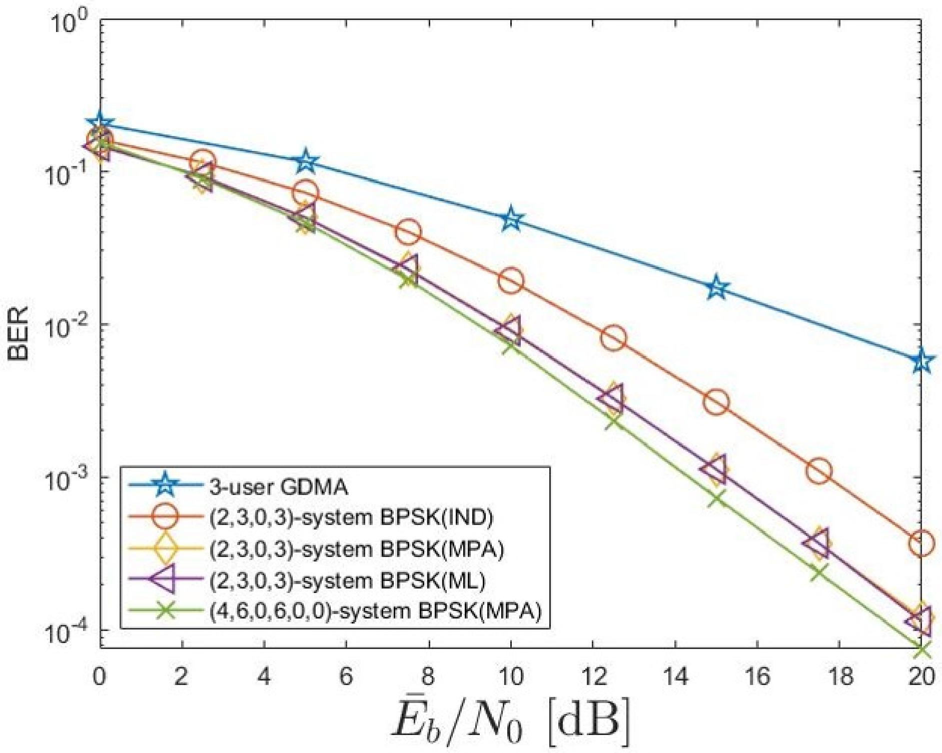

Figure 3.6: BER comparison of (2,3,0,3) and (4,6,0,6,0,0) BPSK modulation.

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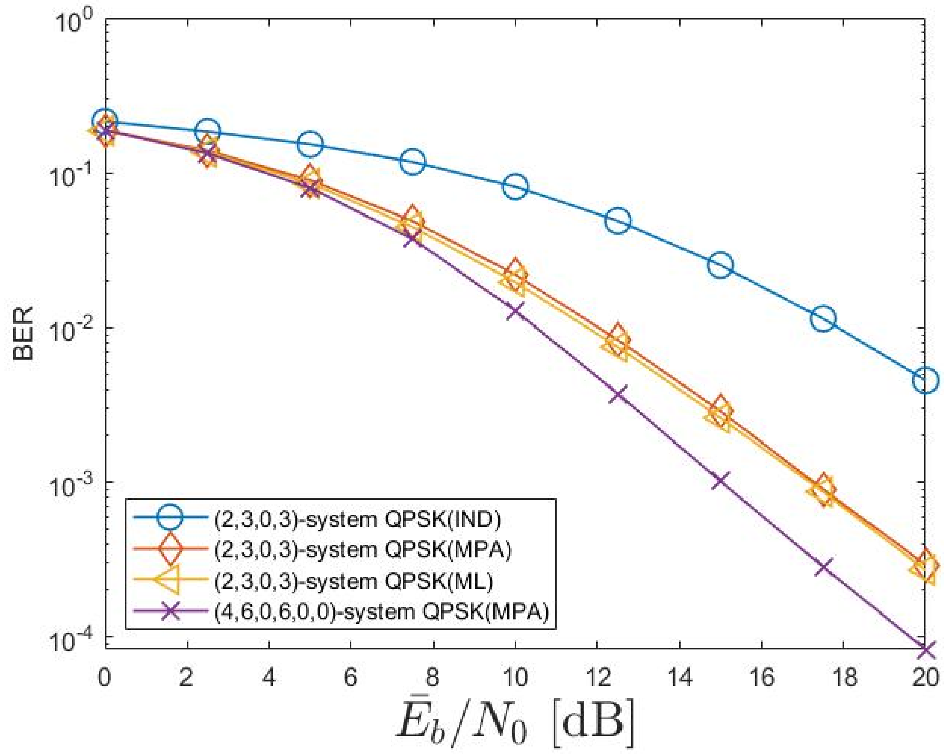

Figure 3.7: BER comparison of (2,3,0,3) and (4,6,0,6,0,0) QPSK modulation.

In terms of BER performance, the (4,6,0,6,0,0) LDS-CDMA system and (2,3,0,3) HDS-CDMA system significantly outperform the (1,3,3) GDMA system. This difference is primarily attributed to the varying diversity values of these systems.

In Fig. 3.6 and Fig. 3.7, one may observe under (E_b/N_0) of 20 dB, the $(2,3,0,3)$ HDS-CDMA system achieves BER that is approximately 1.3 times under BPSK or 3.2 times under QPSK compared to the $(4,6,0,6,0,0)$ LDS-CDMA system. Despite having the same transmission efficiency as the $(4,6,0,6,0,0)$ LDS-CDMA system, the $(2,3,0,3)$ HDS-CDMA system needs less resources and delivers slightly inferior BER performance. It is also not surprising to find BER performance of IND falls between that of (1,3,3) GDMA and (2,3,0,3) HDS-CDMA applying ML or MPA but with least complexity.

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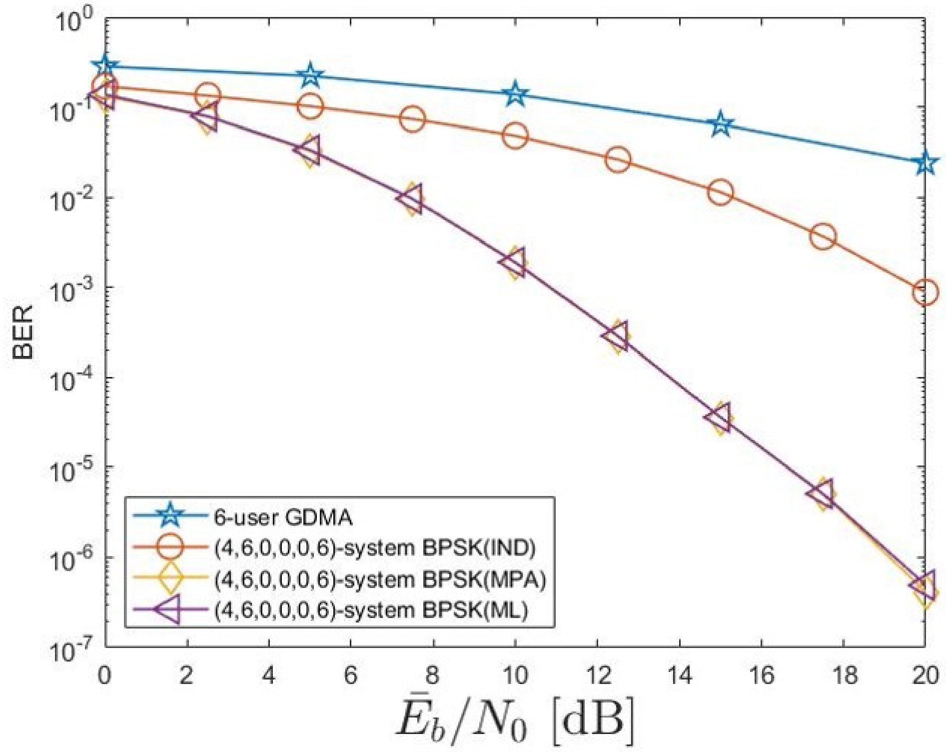

Figure 3.8: Simulated BER curves for (4,6,0,0,0,6) systems (BPSK)

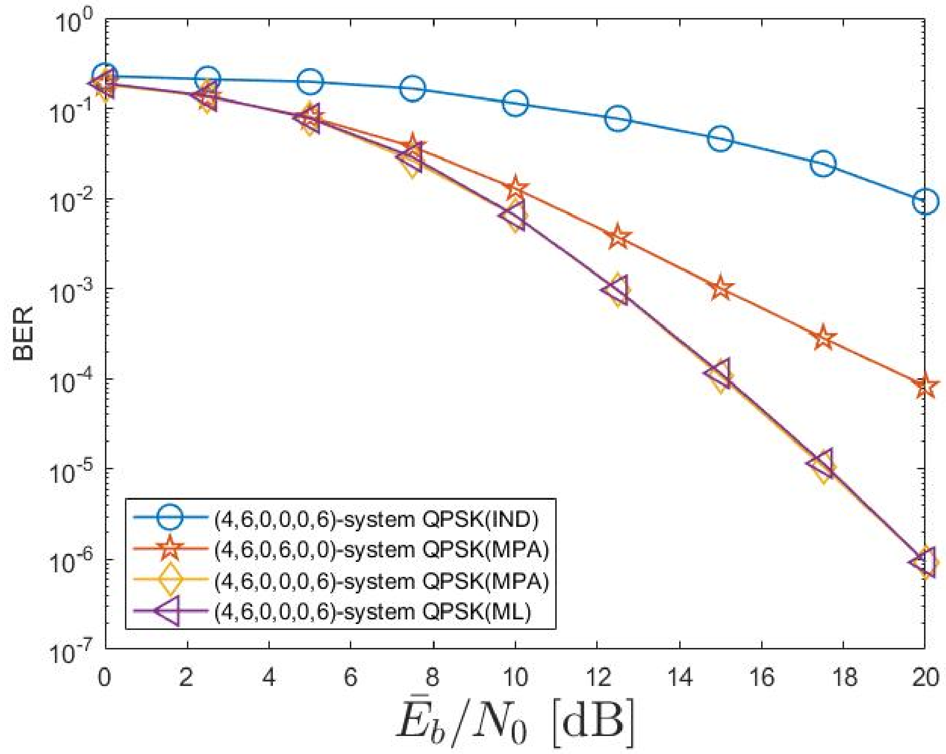

Figure 3.9: Simulated BER curves for (4,6,0,0,0,6) systems (QPSK)

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Consider a $(4,6,0,0,0,6)$ HDS-CDMA system illustrated in Fig. 3.3, in which every resource is shared among 6 users. It may be viewed as consisting of four interconnected 6-user GDMA systems, with each user experiencing a diversity of 4. Fig. 3.8 and Fig. 3.9 show the BER performance using BPSK and QPSK, considering all IND, MPA, and MAP detection methods. With the $(4,6,0,0,0,6)$ system at 20 dB, the BER attained is approximately $\frac{1}{100}$ compared to that achieved by the $(4,6,0,6,0,0)$ LDS-CDMA system.

3.4 Error Correcting Code and OFDM Systems

In this section, we utilize LDPC in GDMA and HDS-CDMA systems to enhance system reliability [11] [12]. Specifically, we apply identity LDPC codes to each user and assess their effectiveness in error correction. Furthermore, we integrate them into the OFDM system to combat multipath and interference effects. The block diagram of the coded OFDM system is depicted in Fig. 3.10.

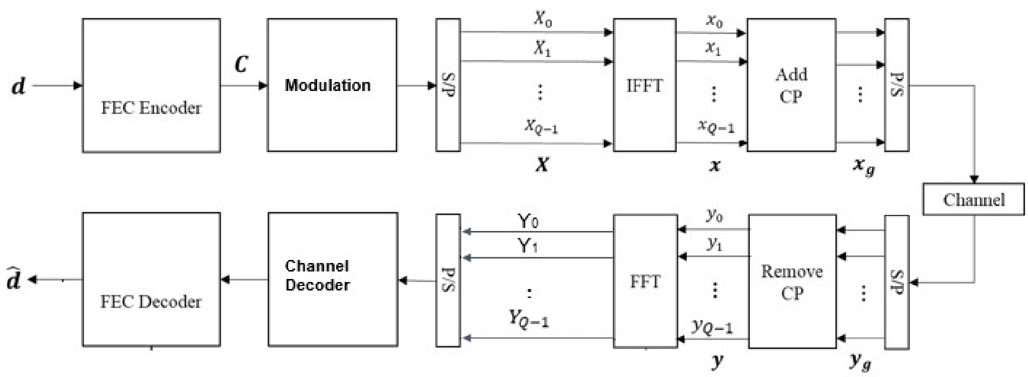

Figure 3.10: The framework for Coded OFDM system

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3.4.1 Brief Reviews of LDPC and OFDM

Low-Density Parity-Check (LDPC)

Low-Density Parity-Check (LDPC) codes are a type of error-correcting code used to enhance data reliability in digital communication systems. Invented by Robert G. Gallager in the 1960s, LDPC codes are known for their ability to approach the Shannon limit, which represents the maximum possible efficiency of error-correcting codes. LDPC codes achieve this by using a sparse bipartite graph to represent the parity-check matrix, enabling iterative decoding algorithms such as the belief propagation algorithm to efficiently detect and correct errors. The key advantage of LDPC codes lies in their excellent error-correction performance, which is achieved with relatively low computational complexity, making them highly suitable for modern communication standards like 5G, Wi-Fi, and satellite communications.

Orthogonal Frequency-Division Multiplexing (OFDM)

Orthogonal Frequency-Division Multiplexing (OFDM) is a multi-carrier modulation technique widely used in modern wireless communication systems. It works by dividing a high-rate data stream into multiple lower-rate streams, which are then transmitted simultaneously over a large number of closely spaced subcarriers. The orthogonality of the subcarriers ensures that they do not interfere with each other, even though they overlap in the frequency domain. This efficient utilization of the spectrum makes OFDM highly resilient to frequency-selective fading and multipath interference, which are common issues in wireless environments. Moreover, OFDM's ability to support high data rates and its robustness against channel impairments have made it a fundamental technology in standards

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such as LTE, Wi-Fi, and digital broadcasting systems.

3.4.2 OFDM-GDMA and OFDM-HDS-CDMA

By applying GDMA and HDS-CDMA to each subcarrier of the OFDM system, we can obtain OFDM-GDMA and OFDM-HDS-CDMA, respectively.

The detection approach can be adapted for orthogonal frequency-division multiplexing. For each $j \in \{0, 1, \ldots, N/(mQ)-1\}$, the subblock $[x_u(jQ+1), x_u(jQ+2), \ldots, x_u((j+1)Q)]$ is transformed into time-domain signals using a Q-point inverse fast Fourier transform (IFFT). To avoid intersymbol interference (ISI) and intercarrier interference (ICI), a cyclic prefix (CP) is added. At the receiver, after removing the CP and performing the Q-point FFT, the signal is processed in the frequency domain for GDMA detection.

Let $H_u(q)$ represent the channel coefficient for user u on the q-th subchannel, where $q \in \{1, \ldots, Q\}$. The received signal component for the q-th subchannel in the frequency domain is expressed as:

$$r(jQ+q) = \sum_{w=1}^{U} H_w(q) x_w(jQ+q) + W(jQ+q) \tag{3.31}$$

where $W(jQ+q)$ denotes the additive white Gaussian noise (AWGN).

3.4.3 Simulation Results

Consider a scenario where a (6,3) regular low-density parity-check code with parameters (1008,504) is used to encode 504-bit message. The resulting 1008 bits are mapped onto 63 orthogonal frequency-division multiplexing symbols, each containing a cyclic

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prefix of length 4 and a subcarrier count Q of 16. The channel model involves a 5-tap Rayleigh fading channel with exponentially decaying power in the taps for every user.

The simulation results for block error rates (BLER) of coded OFDM-GDMA, 2-resource OFDM-HDS-CDMA, and 4-resource OFDM-HDS-CDMA are illustrated in Fig. 3.11, Fig. 3.12, and Fig. 3.13, respectively.

Fig. 3.11 presents the BLER performance of OFDM-GDMA when U users simultaneously occupy the same resource, with U ranging from 1 to 8. Fig. 3.12 displays the BLER performance of $(2, U, 0, U)$ OFDM-HDS-CDMA under the condition where U users concurrently occupy two resources, again with U spanning from 1 to 8. Finally, Fig. 3.13 depicts the BLER performance of $(4, U, 0, 0, 0, U)$ OFDM-HDS-CDMA for the same range of U users, but with four resources being utilized simultaneously. Fig. 3.14 demonstates BLER curves for 6-user coded OFDM system under different usage of resources. We obeserved that the greater the number of users of diversity 2, the better BLER it has on average. The advantage of HDS-CDMA schemes can be the need for fewer resources or improved error performance.

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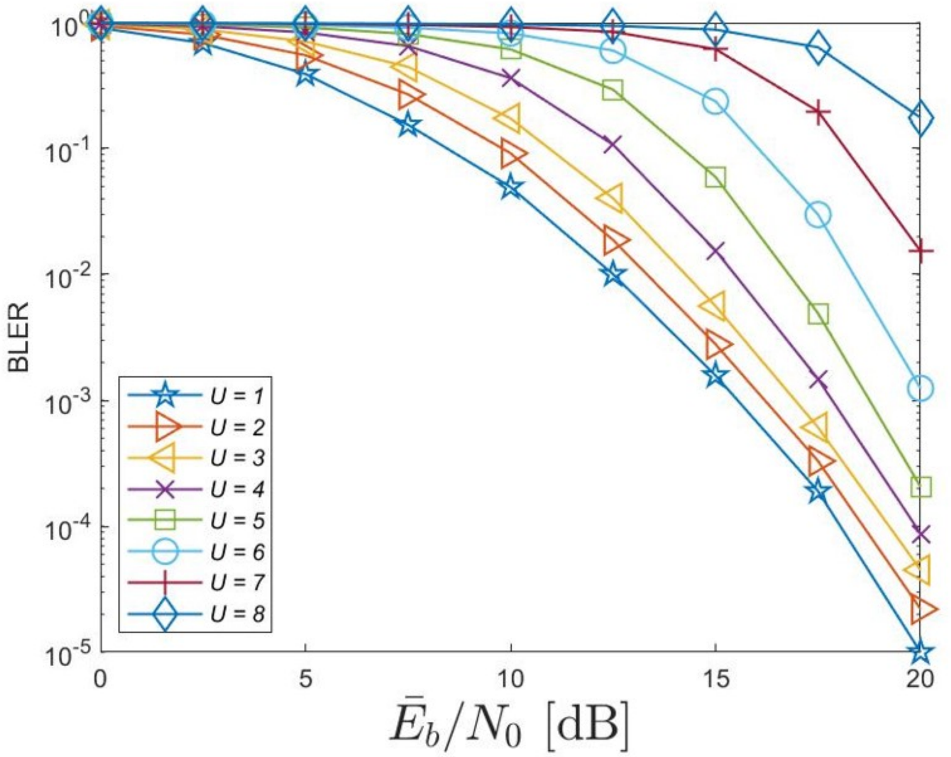

Figure 3.11: Simulated BLER curves for coded OFDM-GDMA over independent 5-tap Rayleigh fading channels based on BPSK modulation.

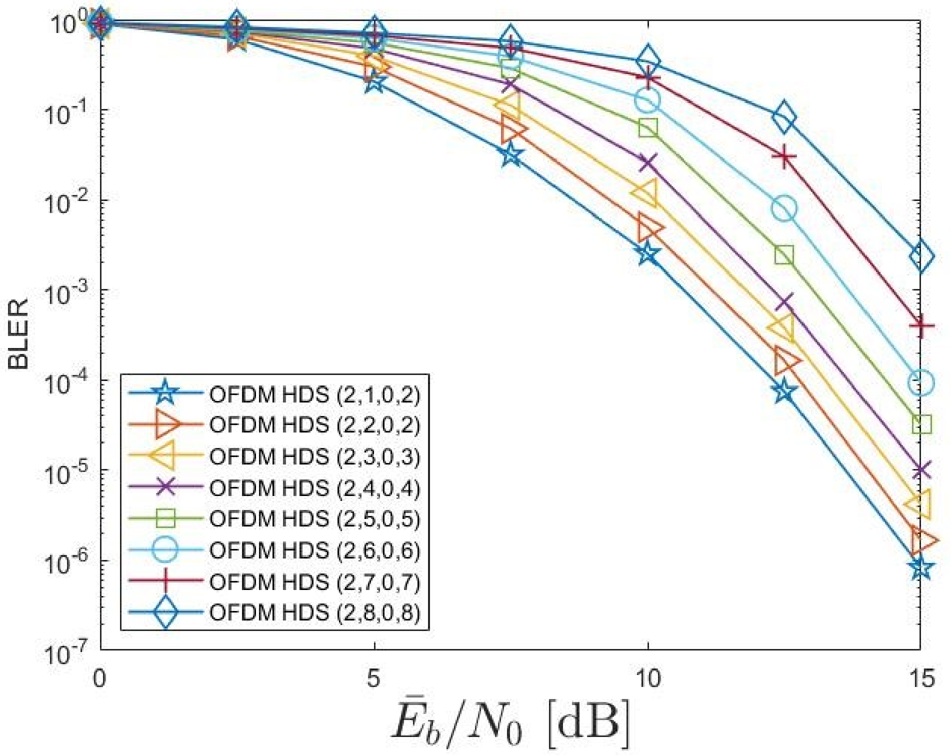

Figure 3.12: Simulated BLER curves for 2-resource coded OFDM-HDS-CDMA over independent 5-tap Rayleigh fading channels based on BPSK modulation.

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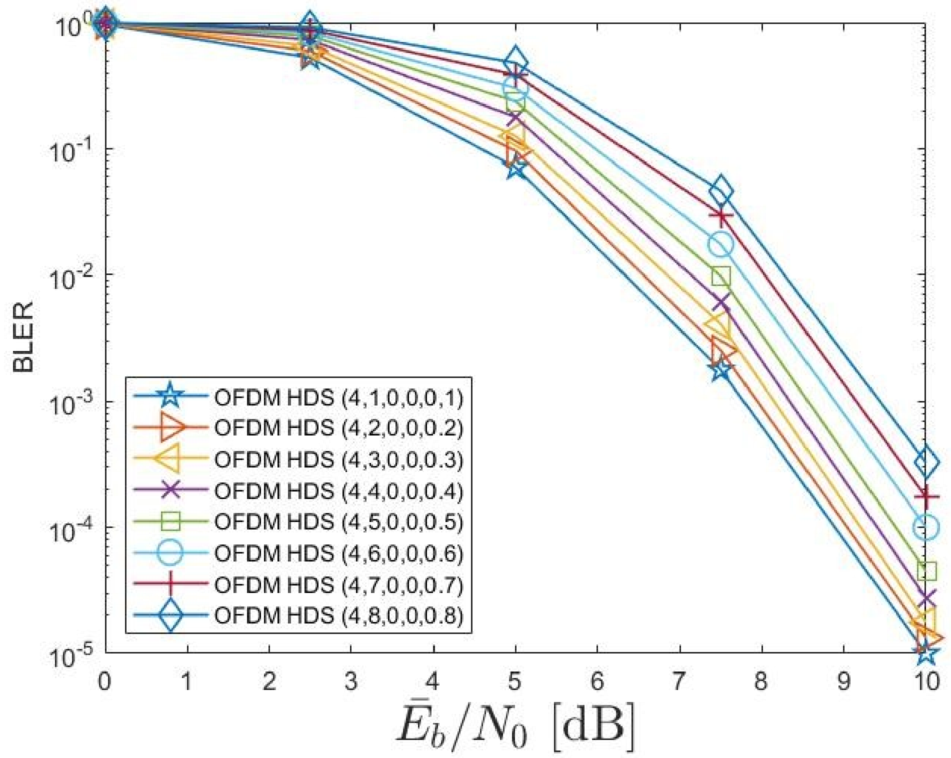

Figure 3.13: Simulated BLER curves for 4-resource coded OFDM-HDS-CDMA over independent 5-tap Rayleigh fading channels based on BPSK modulation.

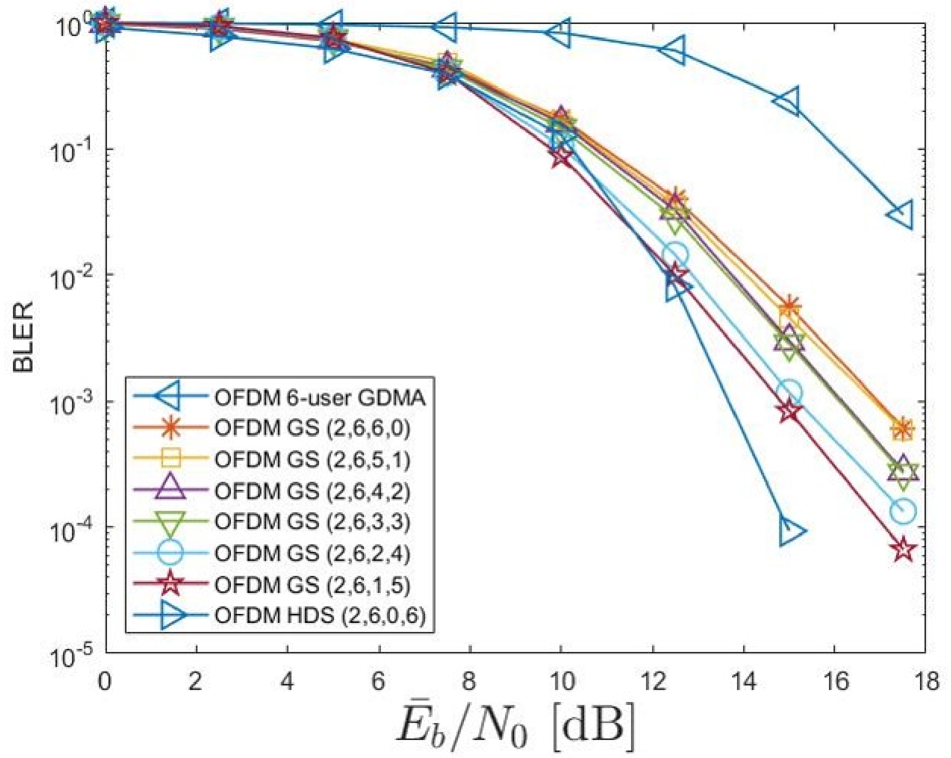

Figure 3.14: Simulated BLER curves for 6-user coded OFDM system under different usage of resources.

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Chapter 4 GDMA and HDS-CDMA Assisted Random Access Scheme

Random access refers to the process of accessing the transmission medium at any given time. Random access generally implies device competition, as opposed to pre-scheduled transmission media, leading to contention for shared transmission space. Due to limited resources, it is impractical for every device to continuously occupy resources. Therefore, a random access mechanism is employed when a communication device needs to access resources. Random access is utilized in various scenarios, such as uplink synchronization between UE and eNB, the random access procedure and the protocol. In this chapter, we will focus on uplink synchronization and apply GDMA and HDS-CDMA to the random access channel using the ALOHA protocol, and then compare the throughput performance of different systems.

4.1 Brief Review of Random Access Communication

Massive Machine Type Communication (mMTC) is designed to enable the connectivity of a vast number of users with sporadic transmission needs. The ALOHA system

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is a random access protocol which facilitates the realization of these sporadic transmissions. In a basic ALOHA system, collisions between multiple transmissions are considered failures. Various ALOHA system variations have been proposed in the literature. Pure ALOHA[16], also known as unslotted ALOHA, does not require time synchronization. In contrast, slotted ALOHA [5] [17] utilizes time synchronization to prevent collisions between packets from different users in adjacent time slots. CSMA/CA [18] monitors the channel to avoid collisions. Additionally, CRDSA [19] mitigates the effects of collisions by randomly assigning repeated packets to different slots.

A multiple access system can enhance the base station's ability to resolve collisions from transmissions by several devices. Therefore, a modified ALOHA system incorporating the multiple access concept can achieve higher throughput compared to its collision-free counterpart.

Suppose that N resources are available to a multiple access system, where a resource can be a frequency band, a time slot, or a signature. In the well-known CDMA system, every user would be assigned a unique spreading sequence (i.e., a unique signature). Therefore, for CDMA, the simultaneous transmission of U users requires $N = U$ resources. Non-orthogonal multiple access (NOMA) [2] [3] is a promising approach for future communication systems, with various schemes proposed. In power-domain NOMA (PD-NOMA), when U users simultaneously use a single resource ($N = 1$), they can be separated by the unique power level of each user. GDMA is another NOMA scheme that allows U users to share a resource ($N = 1$) by utilizing the unique channel gain corresponding to each user for separation. The GDMA system using OFDM modulation is termed OFDM-GDMA.

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PD-NOMA [20] has been applied to ALOHA systems to combat collisions, resulting in a system termed PD-NOMA-ALOHA, which introduces a higher probability of successful decoding during packet collisions compared to the collision-free ALOHA system.

OFDM-GDMA has been integrated into the ALOHA system, as demonstrated in [11], and is denoted by OFDM-GDMA-ALOHA. Moreover, the incorporation of OFDM-HDS-CDMA into the ALOHA system has been demonstrated in [12], and this combined system is referred to as OFDM-HDS-ALOHA. In this thesis,the design is to improve system traffic for the two-resource ALOHA system and extend this approach to the multiple-resource case.

4.2 Pure ALOHA and Slotted ALOHA

In the realm of wireless communication networks, effective and efficient medium access control (MAC) protocols are essential for managing how data packets are transmitted and received. Among the earliest and most foundational MAC protocols are Unslotted ALOHA and Slotted ALOHA, both developed to address the challenges of network collisions and optimize throughput.

Unslotted ALOHA is a pioneering protocol introduced in the early 1970s by Norman Abramson at the University of Hawaii. It operates on a simple premise: devices can transmit data whenever they have data to send. However, this simplicity comes at the cost of a high probability of packet collisions, as there is no coordination among devices. When collisions occur, the affected devices must wait for a random period before attempting to retransmit their data. Despite its straightforward approach, the efficiency of Unslotted ALOHA is limited, with a theoretical maximum throughput of about 0.1839.

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To enhance the performance of the ALOHA system, Slotted ALOHA was introduced. This variant improves upon Unslotted ALOHA by dividing the time into discrete slots equal to the packet transmission time. Devices are required to align their transmissions to these slots, effectively reducing the chances of collision. By ensuring that data packets start only at the beginning of a time slot, Slotted ALOHA increases the maximum theoretical throughput to approximately 0.3679. This doubling in efficiency highlights the significance of temporal synchronization in reducing packet collisions and enhancing network performance.Their schematic diagrams are as shown in Fig. 4.1.

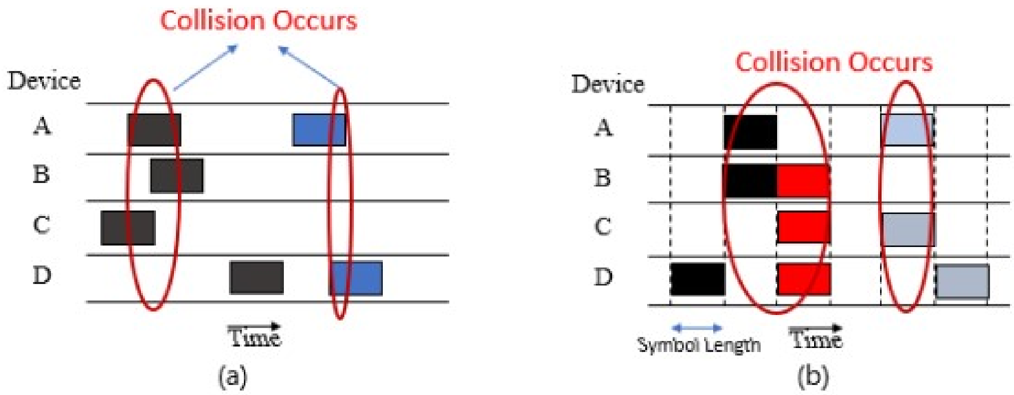

Figure 4.1: (a)An Unslotted ALOHA system. (b)A Slotted ALOHA system.

Both Pure ALOHA and Slotted ALOHA laid the groundwork for more advanced MAC protocols, influencing the development of subsequent communication standards and technologies. Their fundamental principles continue to be relevant in understanding the dynamics of data transmission in modern wireless networks.

4.3 OFDM-GDMA-Q-ALOHA and OFDM-HDS-Q-ALOHA

Incorporating OFDM-GDMA into the slotted ALOHA system results in an efficient random access system, referred to as OFDM-GDMA-ALOHA. HDS-CDMA may be

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viewed as a combination of Q-dependent GDMA systems, where each GDMA is applied to one of the Q resources. By integrating Q-resource OFDM-HDS-CDMA into the slotted ALOHA system, we obtain OFDM-HDS-Q-ALOHA. Given Q available resources, we compare the throughput performance of OFDM-GDMA-Q-ALOHA and OFDM-HDS-Q-ALOHA. For the latter system, every user randomly selects one of the Q resources for packet transmission, after which OFDM-GDMA-ALOHA is applied to that resource.The distinction between a traditional slotted ALOHA system and the slotted OFDM-GDMA-1-ALOHA system is illustrated in Fig. 4.2. Moreover, OFDM-GDMA-2-ALOHA system and OFDM-HDS-Q-ALOHA are illstrated in Fig. 4.3 and Fig. 4.4 respectively.

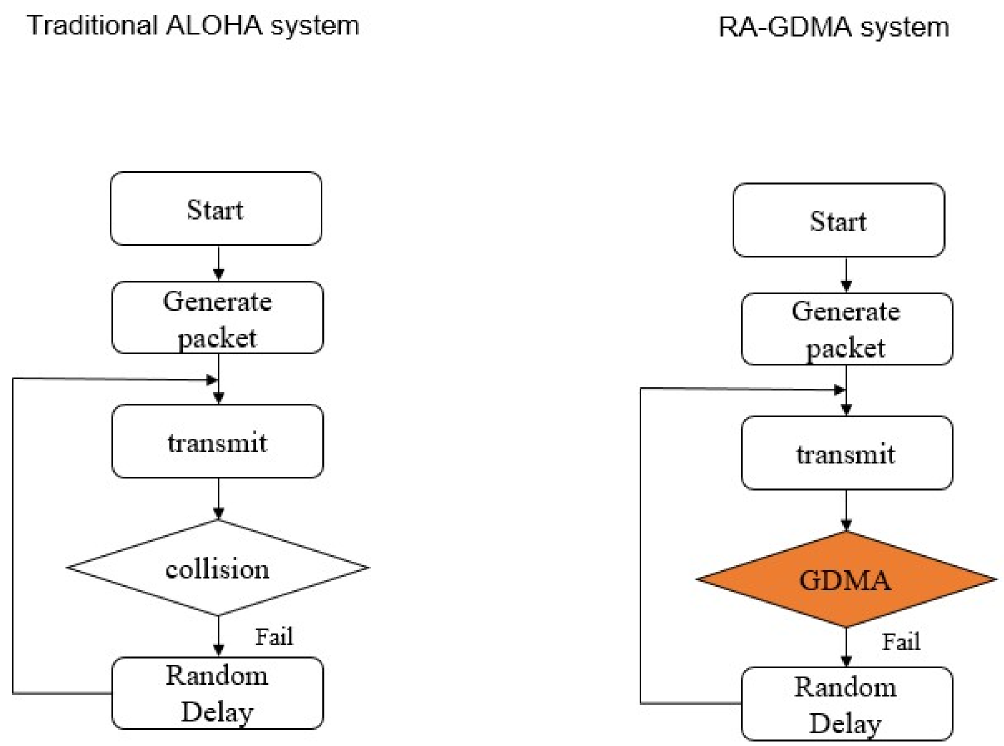

Figure 4.2: (a) A traditional slotted ALOHA system. (b) The slotted OFDM-GDMA-1-ALOHA system

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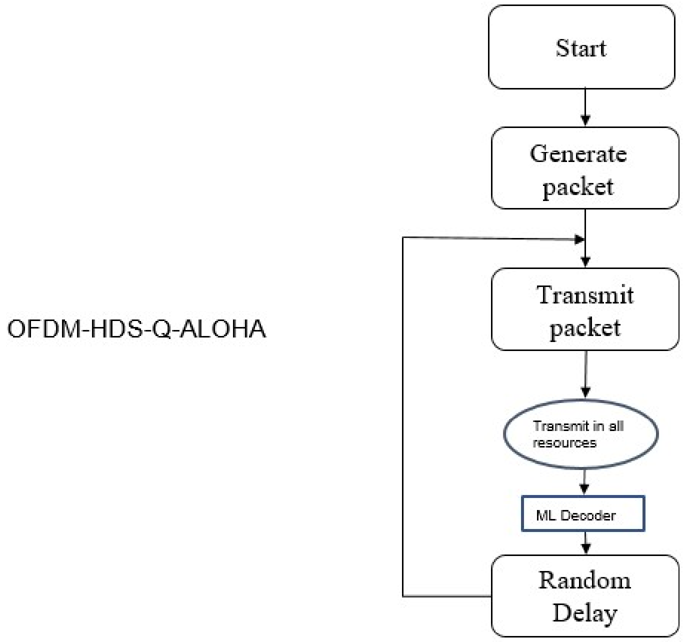

Figure 4.4: (a)Flow chart of a OFDM-HDS-Q-ALOHA system

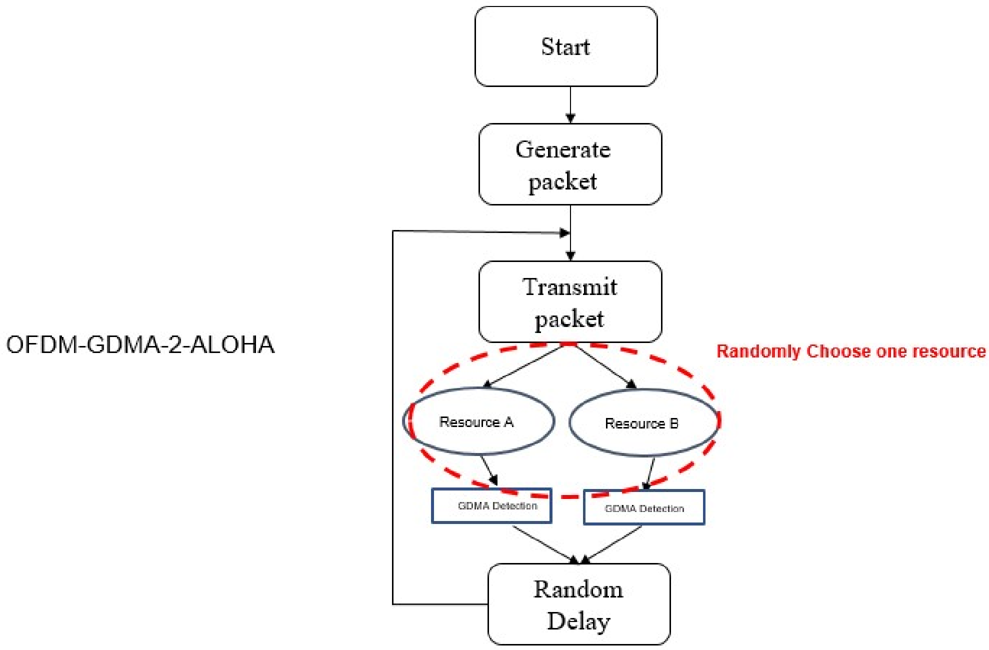

Figure 4.3: (a)Flow chart of a OFDM-GDMA-2-ALOHA system

As the number of devices increases, the likelihood of multiple users transmitting packets in the same frame rises, leading to collisions and failures. The slotted OFDM-GDMA-Q-ALOHA and OFDM-HDS-Q-ALOHA systems aim to address the issue of si-

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multaneous collisions by enabling the detection of multiple users' packets through conventional GDMA techniques.

4.4 Throughput Performances and Complexity Analysis of ALOHA Systems

We first review a way to predict throughput of the conventional unslotted ALOHA and slotted ALOHA systems, and then introduce a way to derive throughput of any slotted ALOHA system.

Assume that in the ALOHA system, packet transmission follows a Poisson process with an average rate of λ packets per second from the transmitter (user) side. When the same packet is simultaneously transmitted to multiple resources in an OFDM-HDS-CDMA based ALOHA system, it is considered as a single packet transmission. The normalized channel traffic, denoted by G, sstands for $G = \lambda \cdot T_p$ packets per time interval, where T_p is the packet transmission time. Set NP as the random variable acting for the total number of packets transmitted within a time slot, where the length of a time slot is set as the duration, T_p. The probability of z transmitted packets appearing within T_p is

$$P(NP = z) = \frac{G^z e^{-G}}{z!} \tag{4.1}$$

In the unslotted ALOHA system, packets are transmitted at any arbitrary time, where users transmit whenever they have data to send. The successful transmission probability refers to the likelihood that no additional packets are transmitted within the vulnerable period. This period lasts for $2T_{ps}$, as depicted in Fig. 4.5. If two or more packets arrive

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within this period, they will collide. In contrast, the slotted ALOHA system transmits packets within predefined time slots, where each vulnerable period lasts T_p.Therefore, we can calculate the throughput of unslotted ALOHA and slotted ALOHA systems.

Figure 4.5: (a)Vulnerable period for an unslotted ALOHA system. (b)Vulnerable period for a slotted ALOHA system.

For unslotted ALOHA system, its throughput $\eta(G)$

$$\begin{aligned}\eta(G) &= G \cdot \text{Pr}\{\text{no collision}\} \\ &= G \cdot \text{Pr}\{0 \text{ transmission in } 2\,T_p\} \\ &= G \cdot \frac{(2G)^0 \cdot e^{-2G}}{0!} \\ &= G \cdot e^{-2G}\end{aligned} \tag{4.2}$$

For conventional slotted ALOHA system, its throughput $\eta(G)$

$$\begin{aligned}\eta(G) &= G \cdot \text{Pr}\{\text{no collision}\} \\ &= G \cdot \text{Pr}\{0 \text{ transmission in } T_p\} \\ &= G \cdot \frac{(G)^0 \cdot e^{-G}}{0!} \\ &= G \cdot e^{-G}\end{aligned} \tag{4.3}$$

Fig. 4.6 shows the throughput of the ALOHA system. In the unslotted ALOHA system, when $G = 0.5$, the highest throughput S is approximately 0.1839. Conversely, in

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the slotted system, when $G = 1$, the maximum throughput S reaches around 0.3679.

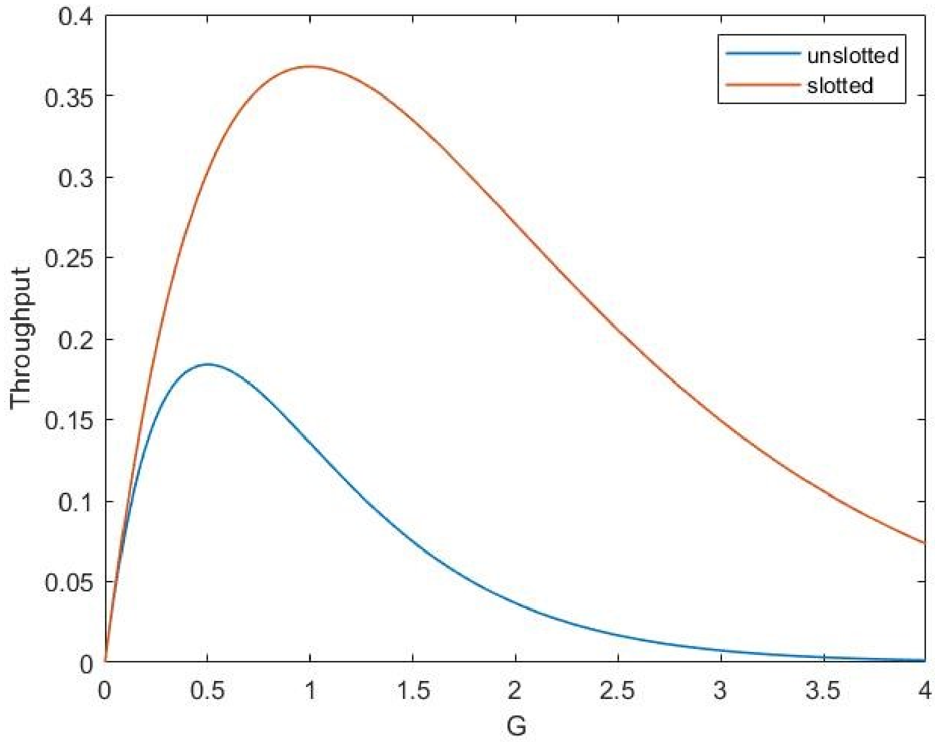

Figure 4.6: Throughput of ALOHA system.

The throughput of any slotted ALOHA system with normalized channel traffic G may be determined by

$$\eta(G) = \sum_{z=1}^{\infty} \sum_{i=1}^{z} i \cdot P(SP = i \mid NP = z) P(NP = z) \tag{4.4}$$

where SP denotes number of successful packets.

For a given $\frac{E_b}{N_0}$, we can evaluate the throughput of any of slotted ALOHA employing any of OFDM-GDMA, 2-resource OFDM-HDS-CDMA, 4-resource OFDM-HDS-CDMA by using equation 4.4. It will be interesting to see the extreme limit of throughput which can be calculated for the condition that every packet is successfully recovered. That means

$$P(SP = z \mid NP = z) = 1. \tag{4.5}$$

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The term

$$\sum_{i=1}^{z} i \cdot P(SP = i \mid NP = z)$$

in equation 4.4 is then equal to z. Then, we have

$$\eta(G) = \sum_{z=1}^{\infty} z \cdot P(NP = z) = G. \tag{4.6}$$

The limit of the throughput $\eta(G)$ is exactly the normalized channel traffic G.

Simulation Results

Based on the same parameters of fading and coding as those used in the coded OFDM systems in Section 3.4.2, we will study the throughput performances of various slotted ALOHA systems. Suppose that the base station provides Q resources for each user to arbitrarily use, where $Q \geq 1$. The user has the following options:

1. Randomly choose one of the Q resources for transmission, denoted as OFDM-GDMA-Q-ALOHA, where $Q \geq 1$;

2. Simultaneously use Q resources by HDS-CDMA, denoted as OFDM-HDS-Q-ALOHA where $m \geq 2$.

1) OFDM-GDMA-Q-ALOHA:

The key operations of the OFDM-ALOHA-Q-ALOHA are listed as follows:

- The number of active users, l, follows the Poisson distribution function and the expected number of active users in a time slot is therefore G.

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- Each user randomly chooses one of the Q resources to access and communicate with the base station. In other words, the expected number of active users in a time slot for each of the Q resources is G/Q.

- Let l_i denote the number of active users who choose the i-th resource. We have $l = \sum_{i=1}^{Q} l_i$.

- A collision in a slot of the i-th channel is resolved by using $(1, l_i, l_i)$ OFDM-GDMA.

Intuitively, the behavior of an OFDM-HDS-Q-ALOHA scheme with normalized channel traffic G is equivalent to Q parallel OFDM-GDMA-1-ALOHA schemes, where the normalized channel traffic of each OFDM-GDMA-1-ALOHA is G/Q. Hence, the throughput, $\eta(G)$, of an OFDM-GDMA-Q-ALOHA scheme with G is $Q \times \eta'(G/Q)$ if the throughput of an OFDM-GDMA-1-ALOHA with normalized channel traffic G/Q is $\eta'(G/Q)$. The maximal throughput of $\eta'(G/Q)$ is G/Q and hence the maximum throughput of $\eta(G)$ is G. For high SNR and small G, the throughput $\eta(G)$ can approach G.

2) OFDM-HDS-Q-ALOHA:

As stated in Section 3.4.3, OFDM-HDS-CDMA can provide better BLER performances compared to OFDM-GDMA. Suppose that there are Q resources available to each user. OFDM-HDS-Q-ALOHA operates as follows:

- The number of active users, l, follows the Poisson distribution function and the expected number of active users is G.

- Each user accesses all the Q resources and transmits the same packet using all the Q resources.

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- If the number of active users is l, the collision is resolved by using Q-resource HDS-CDMA.

The maximal throughput $\eta(G)$ for OFDM-HDS-Q-ALOHA is also G.

3) Simulation for the throughput performances:

In the simulation for the throughput performances, the parameters of fading, coding, and OFDM are the same as those used in Section 3.4.2. We show the throughput of OFDM-GDMA-1-ALOHA, OFDM-GDMA-2-ALOHA, and OFDM-HDS-2-CDMA-ALOHA for $E_b/N_0 = 20$ dB in Fig. 4.7 and for $E_b/N_0 = 5$ dB in Fig. 4.8, respectively, where simulated throughput performances and calculated throughput performances using equation 4.4 with $P(SP = i|NP = z)$ obtained through simulated BLER are used. Note that in Fig. 3.11, Fig. 3.12, and Fig. 3.13 respectively, we only provide the simulated curves of up to 8 collided packets. In the simulation and calculation using (3), we consider the collided packets up to 15. When there are more than 15 collided packets, the system will not be able to resolve the collisions and hence will declare a failure.

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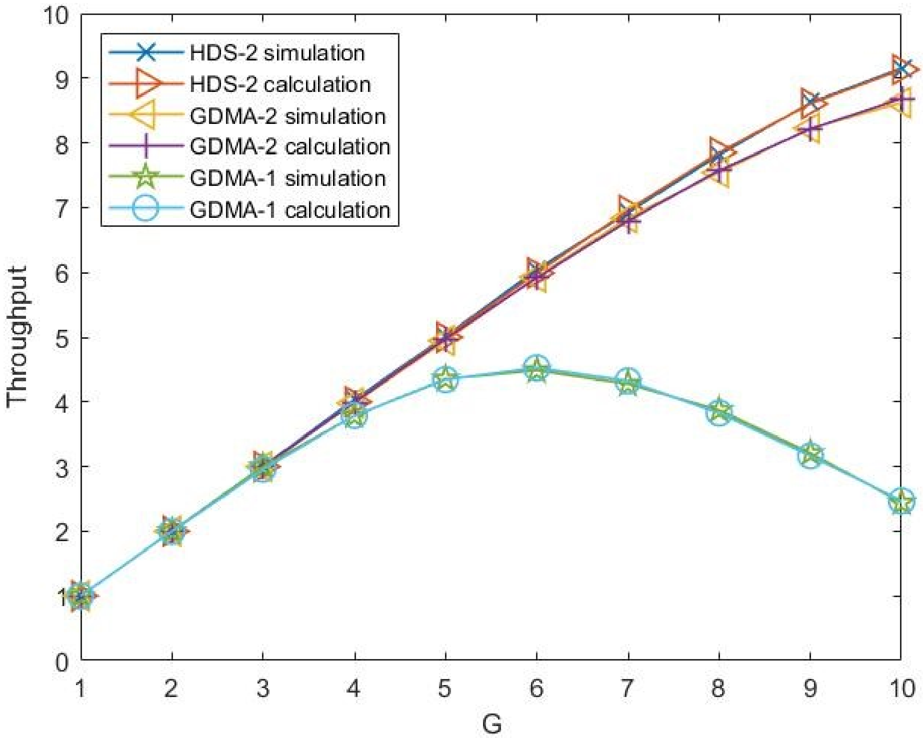

Figure 4.7: Simulation and theoretical calculation of BPSK modulated slotted ALOHA for OFDM-GDMA-1-ALOHA, OFDM-GDMA-2-ALOHA, OFDM-HDS-2-ALOHA. SNR = 20 dB.

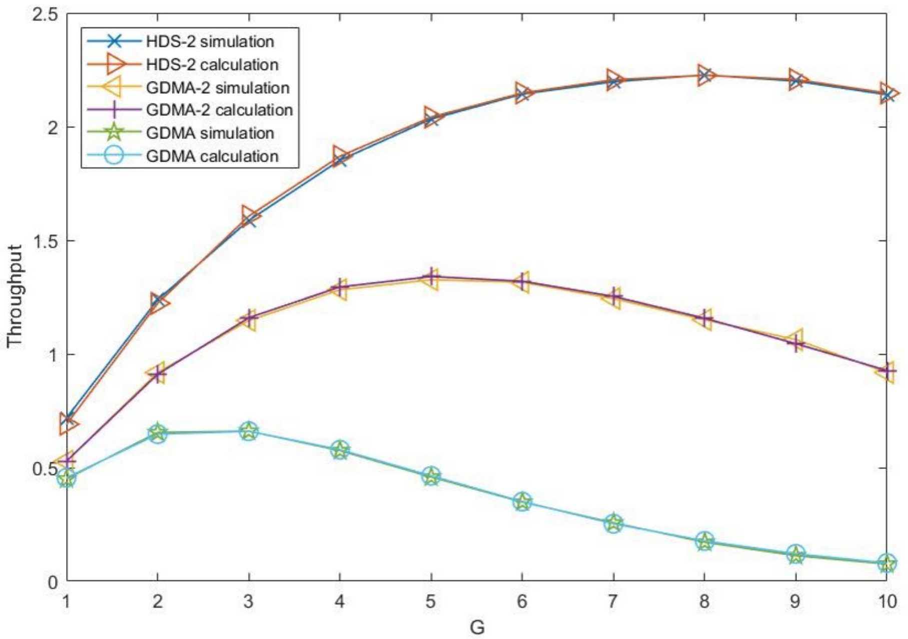

Figure 4.8: Simulation and theoretical calculation of BPSK modulated slotted ALOHA for OFDM-GDMA-1-ALOHA, OFDM-GDMA-2-ALOHA, OFDM-HDS-2-ALOHA. SNR = 5 dB.

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From Fig. 4.7 and Fig. 4.8, it is evident that the throughput of either OFDM-GDMA-2-ALOHA or OFDM-HDS-2-ALOHA is better than that of OFDM-GDMA-1-ALOHA since the former uses two resources and the latter uses only one resource. Fig. 4.7 shows that the advantage in throughput of OFDM-HDS-2-ALOHA over OFDM-GDMA-2-ALOHA is not significant, while Fig. 4.8 shows that the throughput of OFDM-HDS-2-ALOHA is significantly better than that of OFDM-GDMA-2-ALOHA. At high SNR such as $E_b/N_0 = 20$ dB, the success rate, $SP = 1 - \text{BLER}$, for OFDM-GDMA is more than 0.98 for collisions of up to 7 packets. Even though OFDM-HDS-2-CDMA has a better success rate, which is much closer to 1, the small advantage of SP cannot reflect in the gain of throughput. At low SNR such as $E_b/N_0 = 5$ dB, the success rate, $SP = 1 - \text{BLER}$, for OFDM-GDMA is significantly lower than that of OFDM-HDS-2-CDMA. Hence, the advantage in throughput of OFDM-HDS-2-ALOHA can be clearly observed.

Suppose that 4 resources are available. The comparison of throughput performances for OFDM-GDMA-4-ALOHA, OFDM-HDS-4-ALOHA, and other systems can be observed in Fig. 4.9, Fig. 4.10, and Fig. 4.11, where $E_b/N_0 = 20, 10, 5$ dB respectively.

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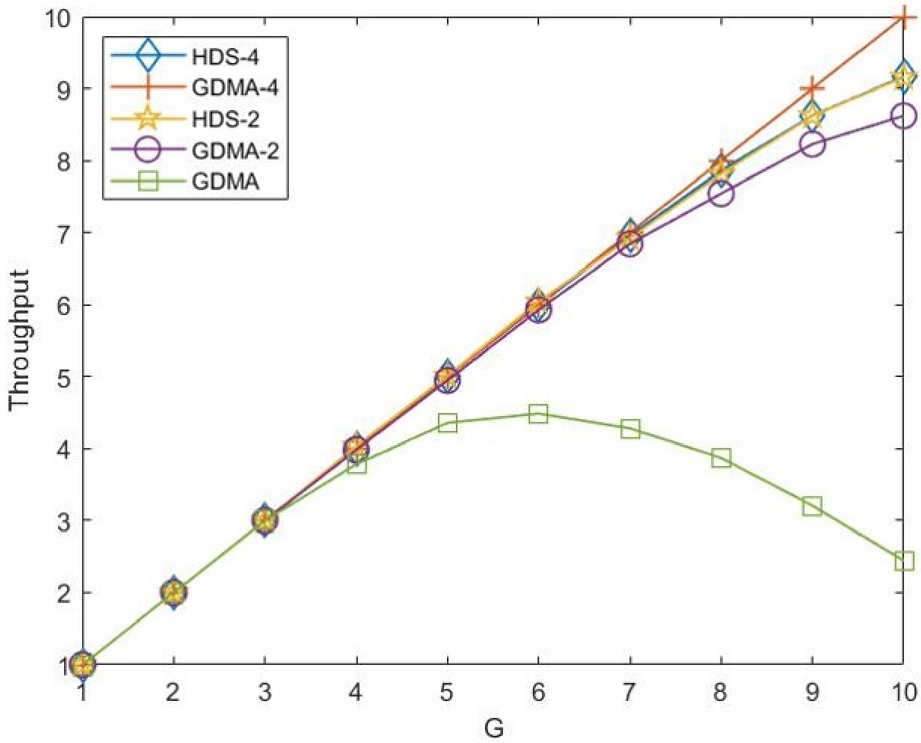

Figure 4.9: Simulation of BPSK modulated slotted ALOHA for various ALOHA systems. SNR = 20 dB.

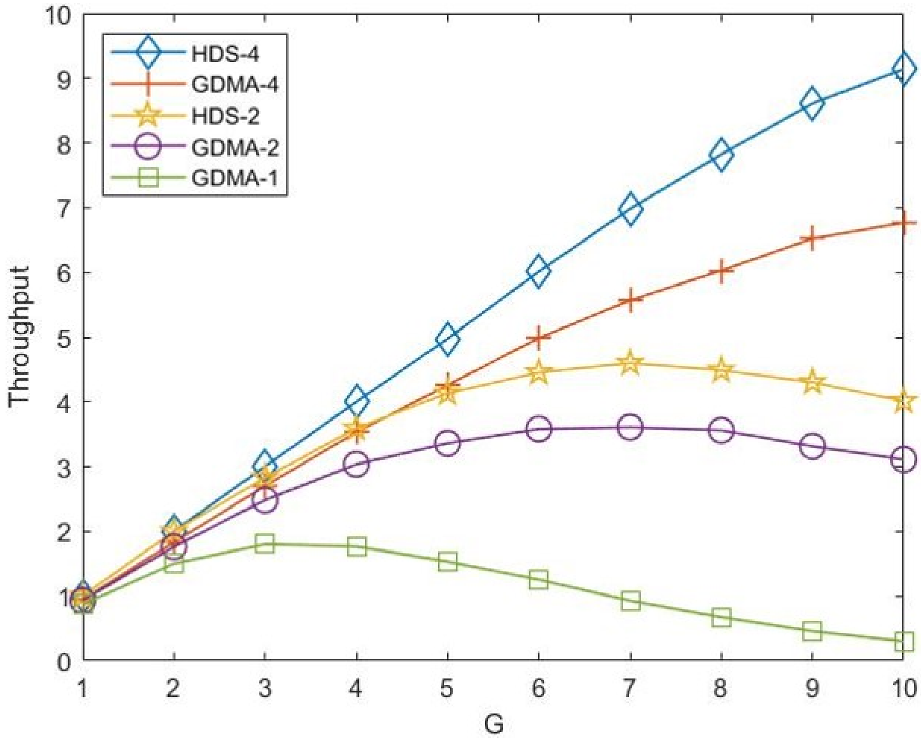

Figure 4.10: Simulation of BPSK modulated slotted ALOHA for various ALOHA systems. SNR = 10 dB.

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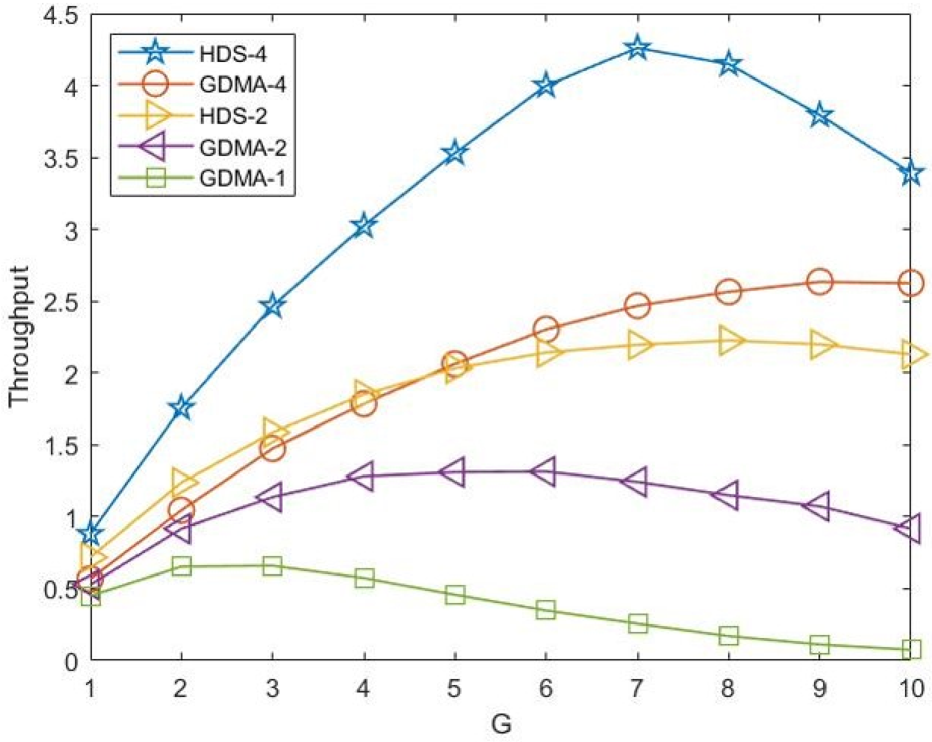

Figure 4.11: Simulation of BPSK modulated slotted ALOHA for various ALOHA systems. SNR = 5 dB.

In our simulation, as speculated before, the behavior of an OFDM-HDS-Q-ALOHA scheme with normalized channel traffic G can be understood as being equivalent to Q independent OFDM-GDMA-1-ALOHA schemes, each with normalized channel traffic G/Q. Consequently, $\eta(G)$ of an OFDM-GDMA-Q-ALOHA scheme with G is $Q \times \eta'(G/Q)$, provided that the throughput of a single OFDM-GDMA-1-ALOHA scheme with normalized channel traffic G/Q is $\eta'(G/Q)$. The maximum throughput of $\eta'(G/Q)$ is G/Q, so the maximum throughput of $\eta(G)$ is G. When the signal-to-noise ratio (SNR) is high, the throughput $\eta(G)$ can closely approach G.

It is observed that OFDM-HDS-4-ALOHA outperforms OFDM-GDMA-4-ALOHA for $1 \leq G \leq 10$. However, with high SNR, as G approaches 10, the advantage of OFDM-HDS-4-ALOHA diminishes. This phenomenon can be attributed to the limitation of a

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maximum of 15 collided users. Under high traffic conditions, the likelihood of having more than 15 collided users is significantly higher in an OFDM-HDS-4-ALOHA system compared to an OFDM-GDMA-4-ALOHA system. Note that OFDM-HDS-4-ALOHA, OFDM-HDS-2-ALOHA, and OFDM-GDMA-4-ALOHA can successfully decode nearly all packets when SNR $= 20$dB, the difference in throughput is due to our limitations.

Complexity

In previous section, we have shown OFDM-HDS-Q-ALOHA is suitable for almost all conditions. The price of using OFDM-HDS-Q-ALOHA is the high complexity of detection. In this section, we will introduce an easy way to measure complexity.

In our simulation, we use ML method to decode. As stated before, in our detection process, we first calculate likelihood functions and then convert to *a posterior* probabilities by Bayesian rule supposing equal *a prior* probabilities. Last, we calculate LLR-values and make decisions based on them. For both OFDM-HDS-Q-ALOHA and OFDM-GDMA-Q-ALOHA, the detection complexity is proportional to the number of likelihood functions calculated.

For OFDM-HDS-Q-ALOHA, suppose there are U users and Q resources. Each user randomly selects one resource to transmit. The number of possible superimposed signals, considering all users, is 2^U. Given that we need to compute 2^U likelihood functions for each resource, the total number of likelihood functions calculated across all resources is:

$$Q \times 2^U$$

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This formula accounts for the computational complexity associated with evaluating the likelihood functions for every possible combination of user transmissions across all resources.

In the context of OFDM-GDMA-Q-ALOHA, suppose there are U users, and each user randomly selects one resource to transmit. For the q-th resource, let U_q denote the number of users transmitting on that resource. The total number of users can then be expressed as:

$$U_1 + U_2 + \cdots + U_Q = U$$

The likelihood function calculation is based on the number of users transmitting on each resource. For each resource q where $U_q \neq 0$, the total number of likelihood functions calculated is given by:

$$2^{U_1} + 2^{U_2} + \cdots + 2^{U_Q}$$

This expression takes into account the exponential growth of the likelihood functions with respect to the number of users on each resource.

An Example : Anticipated Complexity of OFDM-GDMA-Q-ALOHA for $U = 8$ and $Q = 4$

Given $U = 8$ users and $Q = 4$ resources, each user selects one resource randomly and with equal probability. The number of users selecting each resource follows a multinomial distribution.

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Let U_q represent the number of users selecting the q-th resource. The total number of users is:

$$U_1 + U_2 + U_3 + U_4 = 8$$

The complexity for each resource is determined by the number of users on that resource, but if no users select a resource ($U_q = 0$), the complexity for that resource is zero.

To incorporate this, we use an indicator function $\mathbb{I}(U_q > 0)$, which is 1 if $U_q > 0$ and 0 otherwise. The total number of likelihood functions is:

$$\sum_{q=1}^{Q} \mathbb{I}(U_q > 0) \cdot 2^{U_q}$$

We need to calculate the expected complexity:

$$E\left[\sum_{q=1}^{Q} \mathbb{I}(U_q > 0) \cdot 2^{U_q}\right] = \sum_{q=1}^{Q} E[\mathbb{I}(U_q > 0) \cdot 2^{U_q}]$$

By linearity of expectation:

$$E[\mathbb{I}(U_q > 0) \cdot 2^{U_q}] = E[\mathbb{I}(U_q > 0)] \cdot E[2^{U_q} \mid U_q > 0]$$

Given $U_q \sim \text{Binomial}(8, \frac{1}{4})$, we have:

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$$P(U_q > 0) = 1 - P(U_q = 0) = 1 - \left(\frac{3}{4}\right)^8$$

Next, we compute $E[2^{U_q} \mid U_q > 0]$ using the conditional distribution:

$$E[2^{U_q} \mid U_q > 0] = \sum_{k=1}^{8} 2^k \cdot P(U_q = k \mid U_q > 0)$$

where:

$$P(U_q = k \mid U_q > 0) = \frac{P(U_q = k)}{P(U_q > 0)} = \frac{\binom{8}{k}\left(\frac{1}{4}\right)^k\left(\frac{3}{4}\right)^{8-k}}{1 - \left(\frac{3}{4}\right)^8}$$

Finally, we sum the expected complexities across all resources:

$$E\left[\sum_{q=1}^{Q} \mathbb{I}(U_q > 0) \cdot 2^{U_q}\right] = Q \cdot E[2^{U_q} \mid U_q > 0] \cdot P(U_q > 0)$$

Using Python to compute this, we get the expected number of likelihood functions to be calculated.

```
import numpy as np

from scipy.stats import binom


# Parameters

U = 8  # Number of users

Q = 4  # Number of resources

p = 1 / Q  # Probability of selecting any one resource
```


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```
# Probability that U_q > 0
P_Uq_gt_0 = 1 - (3 / 4)**U

# Expected value of 2^U_q given U_q > 0
E_2_Uq_given_Uq_gt_0 = sum(
    2**k * binom.pmf(k, U, p) / P_Uq_gt_0
    for k in range(1, U + 1)
)

# Total expected complexity
total_expected_complexity = Q * E_2_Uq_given_Uq_gt_0 * P_Uq_gt_0

total_expected_complexity
```

- **Exponential Growth**: Both systems exhibit exponential growth in complexity with respect to the number of users, but the factors differ.
  - In OFDM-GDMA-Q-ALOHA, the complexity grows with the sum of exponentials of the number of users on each resource: $2^{U_1} + 2^{U_2} + \cdots + 2^{U_Q}$.
  - In OFDM-HDS-Q-ALOHA, the complexity is a product of the number of resources and the exponential number of user combinations: $Q \times 2^U$.
- **Distribution of Users**:
  - In OFDM-GDMA-Q-ALOHA, the complexity depends on the specific distribution of users across resources. If users are evenly distributed, the complexity

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can be more manageable compared to highly skewed distributions.

– In OFDM-HDS-Q-ALOHA, the complexity is independent of the user distribution and is solely based on the total number of users and resources.

- **Scalability**:

  – OFDM-GDMA-Q-ALOHA might have variable complexity based on user distribution, potentially offering better scalability in certain scenarios.

  – OFDM-HDS-Q-ALOHA has a more predictable but potentially higher complexity due to the direct dependence on the total number of users and resources.

Overall, both systems face exponential complexity challenges, but the exact nature and impact of this complexity differ, making one potentially more suitable than the other depending on the specific application and user distribution.

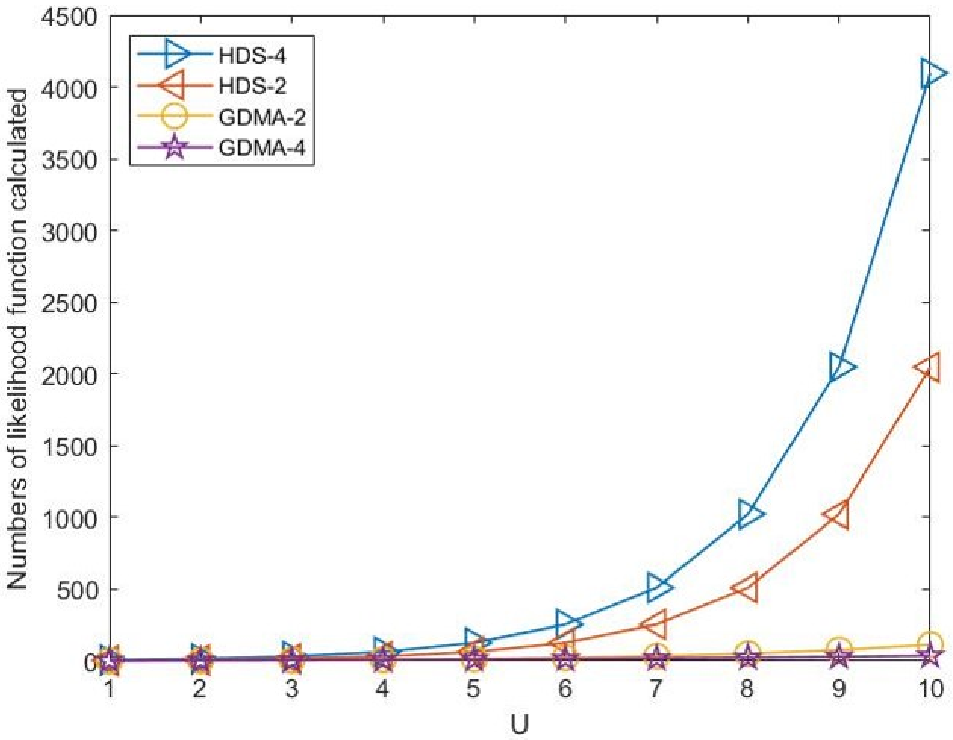

Figure 4.12: Number of likelihood functions calculated by various slotted ALOHA systems

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Fig. 4.12 demonstrates number of likelihood functions calculated by various slotted ALOHA systems, we can observe that the number of likelihood functions calculated by the OFDM-HDS-Q-ALOHA systems increase at a much faster rate than OFDM-GDMA-Q-ALOHA as G grows. However, one needs to notice that there are more processes to derive LLR of each user. For OFDM-HDS-Q-ALOHA, to calculate the Log-Likelihood Ratio (LLR) for each user, we need an additional $(Q-1)\times 2^U$ multiplications, 1 division, and $2^U - 2$ additions. For OFDM-HDS-Q-ALOHA, to calculate the LLR for each user, take users from the q-th resource as an example. We need an additional 1 division and $2^{U_q} - 2$ additions. However, in our simulation, the time required for these operations is significantly less than the time needed to compute the likelihood function. Moreover, due to properties of log function, multiplications and divisions can be implemented by additions and subtractions. If one is interested in reducing complexity, one may further use max approximation.

More Usage of Four Resources

Suppose that 4 resources are available. In the previous section, we analyzed two distinct transmission strategies: one where users randomly select a single resource from four available options, which denotes OFDM-GDMA-4-ALOHA, and another where users utilize all four resources simultaneously, which denotes OFDM-HDS-4-ALOHA. These strategies provided insights into the extremes of resource utilization, ranging from minimal to maximal transmission attempts.

Building upon these foundational scenarios, we now turn our attention to a more nuanced strategy. In this section, we investigate the scenario where users are restricted to transmitting on either the first and second resources or the third and fourth resources. This

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approach introduces a middle ground between the previously studied extremes, allowing us to explore the dynamics of resource partitioning and its impact on overall system performance. Fig. 4.13 is a visualization of this dual-pair transmission strategy, it illustrates the resource usage for up to three users. Moreover, we will denote the case as 2HDS-2 in our simulation graphs.

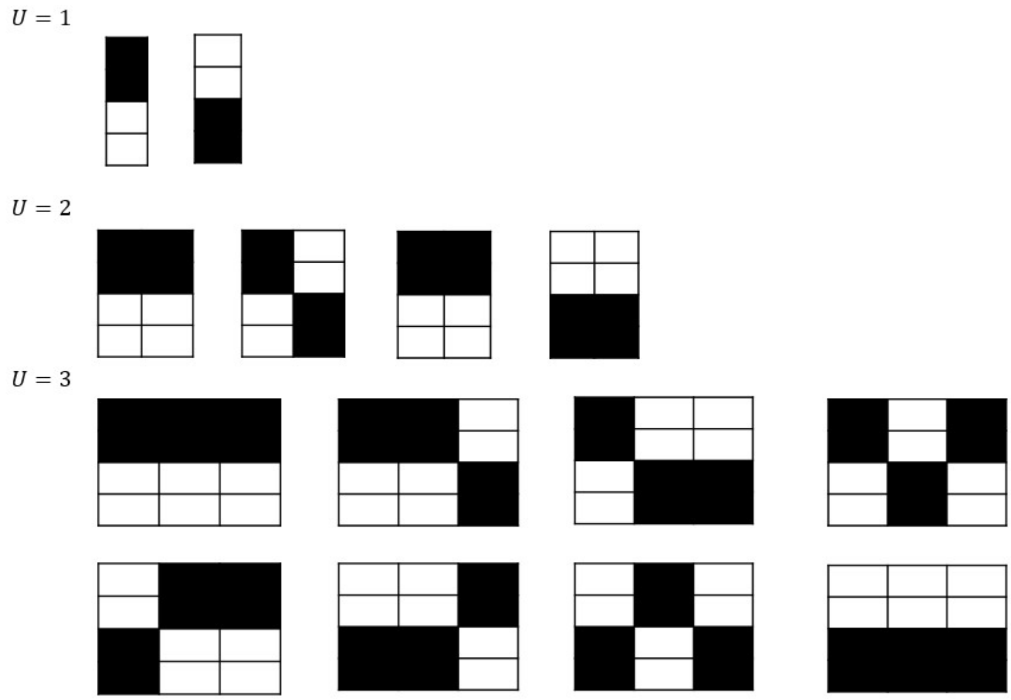

Figure 4.13: Visualization of Dual-Pair Transmission Strategies

To delve into this new transmission strategy, we first outline the rationale behind such a resource division. By limiting users to specific pairs of resources, we aim to strike a balance between resource availability and contention. This partitioning can potentially reduce interference. For larger resource number, some different strategies may improve throughput performance.

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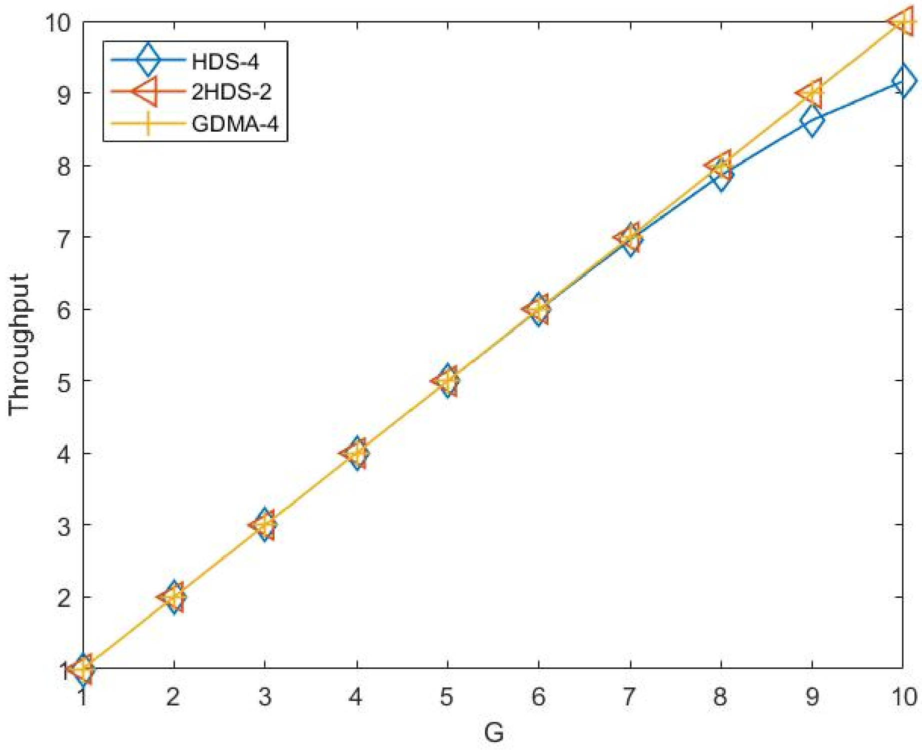

Figure 4.14: Simulation of BPSK modulated slotted ALOHA for the dual-pair strategy and other various ALOHA systems. SNR = 20 dB.

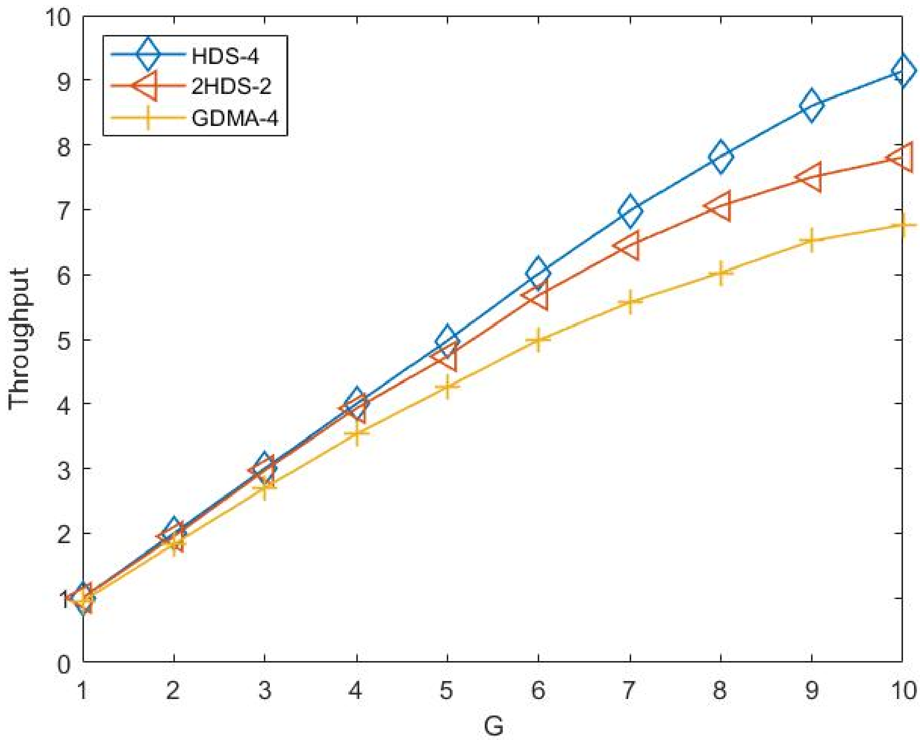

Figure 4.15: Simulation of BPSK modulated slotted ALOHA for the dual-pair strategy and other various ALOHA systems. SNR = 10 dB.

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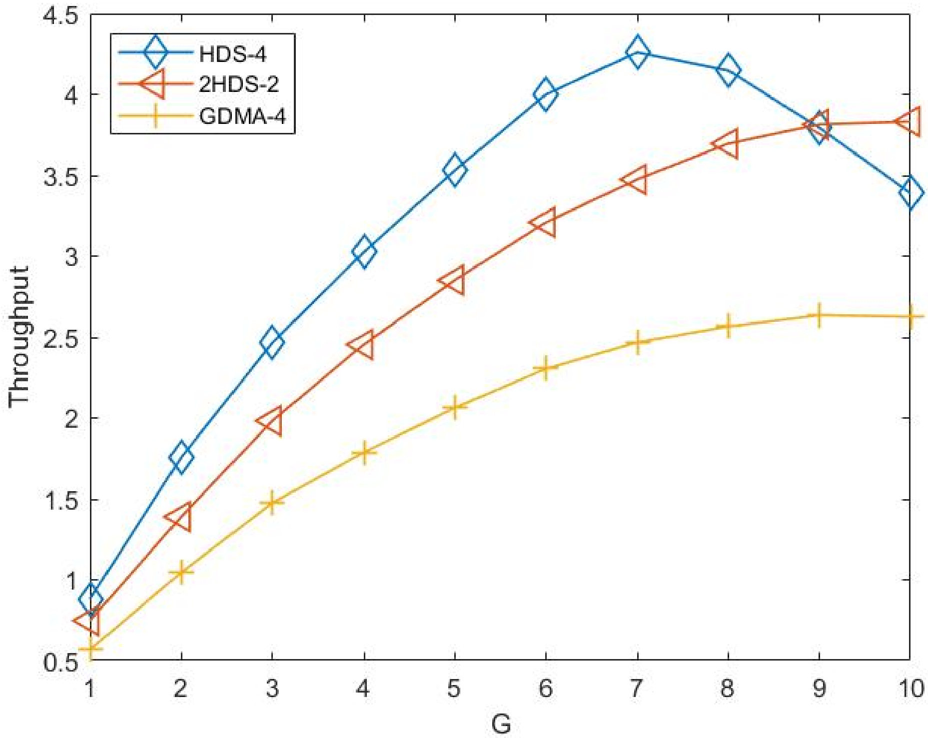

Figure 4.16: Simulation of BPSK modulated slotted ALOHA for the dual-pair strategy and other various ALOHA systems. SNR = 5 dB.

Fig. 4.14 to Fig. 4.16 present the results of resource usage under varying signal-to-noise ratio (SNR) conditions, specifically at 20 dB, 10 dB, and 5 dB, respectively.

Fig. 4.14 illustrates the resource usage at an SNR of 20 dB. Under this condition, we can successfully decode nearly all packets for these three transmission strategies, the difference in throughput is due to our limitations. Fig. 4.15 demonstrates that HDS-4 is capable of successfully decoding near all packets, and the performance of 2HDS-2 lies between HDS-4 and GDMA-4. Fig. 4.16 depicts that 2HDS-2 may have better performance in terms of throughput than HDS-4 at low SNR and high G. The condition reveals significant resource contention and a higher likelihood of collisions across all strategies. Despite the challenging conditions, the dual-pair strategy still demonstrates a relative improvement in managing interference and optimizing resource usage compared to the ran-

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dom and full transmission strategies. It may be interesting to further research on larger resource number and more usage of the resources under different SNR to identify any potential advantages or trade-offs and enlighten the future researches to the broader goal of optimizing resource allocation in multi-user communication environments, ultimately enhancing the performance and reliability of modern wireless networks.

Also, it is not surprising to see that the behavior of a 2HDS-2 scheme with normalized channel traffic G can be understood as being equivalent to 2 independent OFDM-HDS-2-ALOHA schemes, each with normalized channel traffic $G/2$. Consequently, the throughput, $\eta(G)$, of a 2HDS-2 scheme with normalized channel traffic G is $2 \times \eta'(G/2)$, provided that the throughput of a single OFDM-HDS-2-ALOHA scheme with normalized channel traffic $G/2$ is $\eta'(G/2)$.

4.5 Slotted ALOHA with Retransmission Limit

In practical random access networks, a packet discard scheme is commonly used to mitigate network congestion. Under this scheme, a packet is discarded once its retransmission attempts reach a specified limit K. Fig. 4.17 is the block diagram of the retransmission scheme in ALOHA.

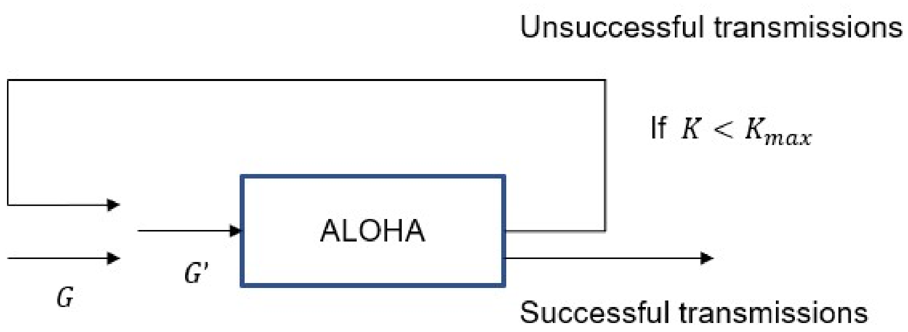

Figure 4.17: The block diagram of the retransmission scheme in ALOHA.

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The process typically works as follows: When a device sends a packet, it waits for an acknowledgment (ACK) from the receiver. If the ACK is not received within a certain time frame, the device assumes a collision has occurred. After detecting a collision, the device waits for a random backoff time before attempting to retransmit, with this randomization helping to minimize the probability of repeated collisions.

In our simulation, we suppose round-trip delay is contained within our slot. Therefore, the backoff window size can start from 0, allowing immediate retransmission attempts if the window size allows. The random backoff time chosen by the device before retransmission is crucial for reducing the likelihood of subsequent collisions, as it ensures that not all devices attempt to retransmit simultaneously.

Simulation Results

In our simulation, the parameters of fading, coding, and OFDM are the same as those used in Section 3.4.2. Moreover, we set the retransmission limit K to 3, meaning that each device is allowed a maximum of three retransmission attempts before discarding the packet. This parameter helps to prevent indefinite retransmission attempts that could lead to congestion and inefficiency in the network. Additionally, we experimented with several different backoff window sizes to analyze their impact on network performance. By varying the initial backoff window sizes, we aimed to identify an optimal range that minimizes collisions while ensuring timely retransmissions. The backoff window sizes were tested across a spectrum, from small windows that encourage quicker retransmissions to larger windows designed to spread out retransmission attempts and reduce the probability of repeated collisions. This comprehensive analysis allowed us to assess how different backoff strategies interact with the transmission limit to influence overall network throughput, de-

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lay, and efficiency.

In our study, we explore two distinct retransmission schemes for ALOHA protocols, each characterized by its unique backoff window size patterns. These schemes are designed to evaluate their impact on the average number of transmission times required for successful packet delivery. The two schemes are as follows:

1. **Scheme 1:** The backoff window sizes are set to $[0, 7]$, $[0, 15]$, and $[0, 31]$. In this scheme, the window size increases progressively with each retransmission attempt, starting from a narrow range and expanding to allow for a larger range of backoff times. This approach aims to reduce the probability of collisions as the number of retransmissions increases.

2. **Scheme 2:** The backoff window sizes are set to $[0, 31]$, $[0, 15]$, and $[0, 7]$. Here, the window size starts with the largest range and decreases after each retransmission attempt. This scheme seeks to quickly resolve collisions by initially allowing a broad range of backoff times and then narrowing the range to focus on resolving persistent collisions.

Considering different success probabilities p_1, p_2, and p_3 for the first, second, and third attempts respectively, the modified offered load G' can be expressed as:

$$G' = G + G(1 - p_1) + G(1 - p_1)(1 - p_2) + G(1 - p_1)(1 - p_2)(1 - p_3)$$

For simplicity, we assume a uniform success probability $\Pr\{\text{success}\}$ across all attempts. In this simplified model, the probability of a packet failing to transmit is $1 -$

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Pr{success}. The portion of traffic that fails and is retransmitted after the first attempt is $G(1 - \Pr\{\text{success}\})$. The probability of failing again in the second attempt is $(1 - \Pr\{\text{success}\})^2$, which corresponds to the traffic transmitted for the third time, $G(1 - \Pr\{\text{success}\})^2$.

This pattern continues, and generally, the amount of traffic transmitted for the i-th time is given by:

$$G \times (1 - \Pr\{\text{success}\})^{i-1}$$

This equation accounts for the total offered load, G', including all retransmissions due to failed attempts.

Consider $K + 1$ as the maximum allowable number of transmissions for a packet (one initial transmission followed by K retransmissions). Since we set K to 3, we get the aggregate traffic G' can be calculated as equation 4.7. In Fig. 4.18, we depict the relationship between G' and G with an exponentially increasing backoff window size to manage packet retransmissions. Specifically, the backoff window sizes are set to progress through the ranges $[0, 7]$, $[0, 15]$, and $[0, 31]$. This approach ensures that the contention window dynamically expands, reducing the probability of repeated collisions and improving overall network performance. By allowing the window size to increase exponentially, we can better accommodate high traffic scenarios and enhance the efficiency of our communication protocol. The theoretical ones are calculated through equation 4.7. The others are the mean aggregate offered traffic in our simulation. The success rate, as shown in Fig. 4.19, in our ALOHA systems plays a crucial role, as our theoretical values are predominantly derived from the success rate.

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$$G' \approx G \times \sum_{i=1}^{K+1} \{1 - \Pr\{\text{success}\}\}^{i-1}. \tag{4.7}$$

The drop rate, as shown in Fig. 4.20, of our system can be described by equation 4.8, but for simplicity, we use equation 4.9 instead, where DR represents the drop rate. In this formula, $\Pr\{\text{Success}\}$ denotes the probability of a successful packet transmission, and $K + 1$ represents the total number of transmission attempts. This relationship highlights how the drop rate is influenced by both the probability of successful packet transmissions and the total number of transmission attempts.

$$\text{DR} = \prod_{i=1}^{K+1} (1 - p_i) \tag{4.8}$$

$$\text{DR} \approx \{1 - \Pr\{\text{Success}\}\}^{K+1} \tag{4.9}$$

The precise throughput, as shown in Fig. 4.21, of our system can be described by the equation 4.11, where $\eta(G)$ represents the throughput as a function of G. In this formula, $\Pr\{\text{Success}\}$ denotes the probability of a successful packet transmission, and $K+1$ stands for the total transmission times. This relationship highlights how the throughput is influenced by both the traffic intensity and the likelihood of successful packet transmissions. The precise throughput, accounting for different success probabilities at each transmission attempt, is given by:

$$\eta(G) = G \left[1 - \prod_{i=1}^{K+1} (1 - p_i) \right] \tag{4.10}$$

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Here, p_i represents the probability of a successful transmission on the i-th attempt, where $i = 1, 2, \ldots, K + 1$.

$$\eta(G) \approx G \left[1 - \{1 - \Pr\{\text{Success}\}\}^{K+1}\right] \tag{4.11}$$

In an ALOHA system, the average number of transmission attempts, as shown in Fig. 4.22, is a critical metric for several reasons. First, it directly impacts the overall system efficiency and throughput. The more attempts required to successfully transmit a packet, the more time and network resources are consumed, which can lead to increased congestion and reduced performance. Additionally, understanding the average transmission attempts helps in designing more robust and efficient retransmission protocols. By analyzing this metric, system designers can optimize the balance between successful transmissions and packet drops, thereby improving the quality of service and reducing latency. Furthermore, the average number of transmission attempts influences the power consumption of devices in the network. Higher transmission attempts typically mean more energy is used, which is a crucial consideration for battery-powered devices in wireless sensor networks or mobile communications. Finally, this metric is essential for predicting and managing the network's capacity. By knowing how many attempts are typically needed, network administrators can better estimate the system's load and make informed decisions on scaling the infrastructure or adjusting operational parameters to ensure reliable and efficient communication.

Derivation of Average Transmission Attempts

To derive the average number of transmission attempts with varying success probabilities for each attempt:

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- Let p_i denote the probability of a successful transmission on the i-th attempt.

- Let $q_i = 1 - p_i$ denote the probability of failure on the i-th attempt.

The expected number of transmission attempts E is calculated as follows:

1. Successful Transmission on the i-th Attempt:

$$\Pr(\text{Success on } i\text{-th attempt}) = \left(\prod_{j=1}^{i-1} q_j\right) p_i$$

The expected attempts for success on the i-th attempt:

$$i \cdot \left(\prod_{j=1}^{i-1} q_j\right) p_i$$

2. Total Expected Attempts for Success:

$$E_{\text{success}} = \sum_{i=1}^{K+1} i \left(\prod_{j=1}^{i-1} q_j\right) p_i$$

3. Attempts in Case of Drop After $K+1$ Attempts:

$$E_{\text{drop}} = (K+1) \cdot \left(\prod_{i=1}^{K+1} q_i\right)$$

4. Total Expected Transmission Attempts:

$$E = E_{\text{success}} + E_{\text{drop}}$$

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$$E = \sum_{i=1}^{K+1} i \left(\prod_{j=1}^{i-1} q_j \right) p_i + (K+1) \left(\prod_{i=1}^{K+1} q_i \right)$$

However, for simplicity, we assume a uniform success probability $\Pr\{\text{success}\}$ across all attempts. We can further derive the closed form under this assumption.

Let:

- $p = \Pr\{\text{Success}\}$ be the probability of a successful packet transmission.
- $q = 1 - p$ be the probability of a failed packet transmission.

The expected number of attempts needed to successfully transmit a packet (including retransmissions) can be derived using the geometric distribution.

Expected Number of Attempts

For each transmission attempt, the packet could either succeed or fail. If the packet fails, it is retransmitted, but only up to K times (i.e., $K + 1$ total attempts). If the packet fails all $K + 1$ attempts, it is dropped.

1. For successful transmissions within $K + 1$ attempts

The probability of success on the i-th attempt is given by:

$$\Pr(\text{Success on } i\text{-th attempt}) = q^{i-1} \cdot p$$

The expected number of attempts if the packet is successfully transmitted within $K+$

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1 attempts is:

$$E_{\text{success}} = \sum_{i=1}^{K+1} i \cdot \Pr(\text{Success on } i\text{-th attempt})$$

$$E_{\text{success}} = \sum_{i=1}^{K+1} i \cdot q^{i-1} \cdot p$$

This can be simplified by recognizing it as a weighted geometric series:

$$E_{\text{success}} = p \sum_{i=1}^{K+1} i \cdot q^{i-1}$$

Using the formula for the sum of an arithmetico-geometric series:

$$\sum_{i=1}^{n} i \cdot q^{i-1} = \frac{1 - (n+1)q^n + nq^{n+1}}{(1-q)^2}$$

For our case, $n = K + 1$:

$$\sum_{i=1}^{K+1} i \cdot q^{i-1} = \frac{1 - (K+2)q^{K+1} + (K+1)q^{K+2}}{(1-q)^2}$$

Substitute this back into E_{success}:

$$E_{\text{success}} = p \cdot \frac{1 - (K+2)q^{K+1} + (K+1)q^{K+2}}{(1-q)^2}$$

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2. For packet drop after $K + 1$ attempts

The probability that all $K + 1$ attempts fail is q^{K+1}, and the packet will be dropped. In this case, the total number of attempts is $K + 1$:

$$E_{\text{drop}} = (K + 1) \cdot q^{K+1}$$

3. Combining both scenarios

The overall expected number of attempts combines both the success and drop scenarios, weighted by their probabilities:

$$E = E_{\text{success}} + E_{\text{drop}}$$

Thus, the overall expected number of transmission attempts is:

$$E \approx p \cdot \frac{1 - (K + 2)q^{K+1} + (K + 1)q^{K+2}}{(1 - q)^2} + (K + 1) \cdot q^{K+1}$$

Also, suppose that 4 resources are available, the performance are shown from Fig. 4.23 to Fig. 4.26. It is seen that OFDM-HDS-4-ALOHA prevails over OFDM-GDMA-4-ALOHA.

Fig. 4.27 and Fig. 4.28 illustrate the relationship between G' and G and success rate with backoff window size $[0, 31]$, $[0, 15]$, and $[0, 7]$. Given that the overall performance improvement is marginal, we present only the success rate for Scheme 2, allowing viewers to derive other performance metrics if needed.

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In conclusion, the choice of backoff window size pattern influences the performance of ALOHA protocols. While Scheme 2 shows a marginal improvement in success rate over Scheme 1, the difference is not substantial under certain conditions. Future research could explore hybrid approaches or adaptive schemes that dynamically adjust the backoff windows based on real-time network conditions to further enhance performance.

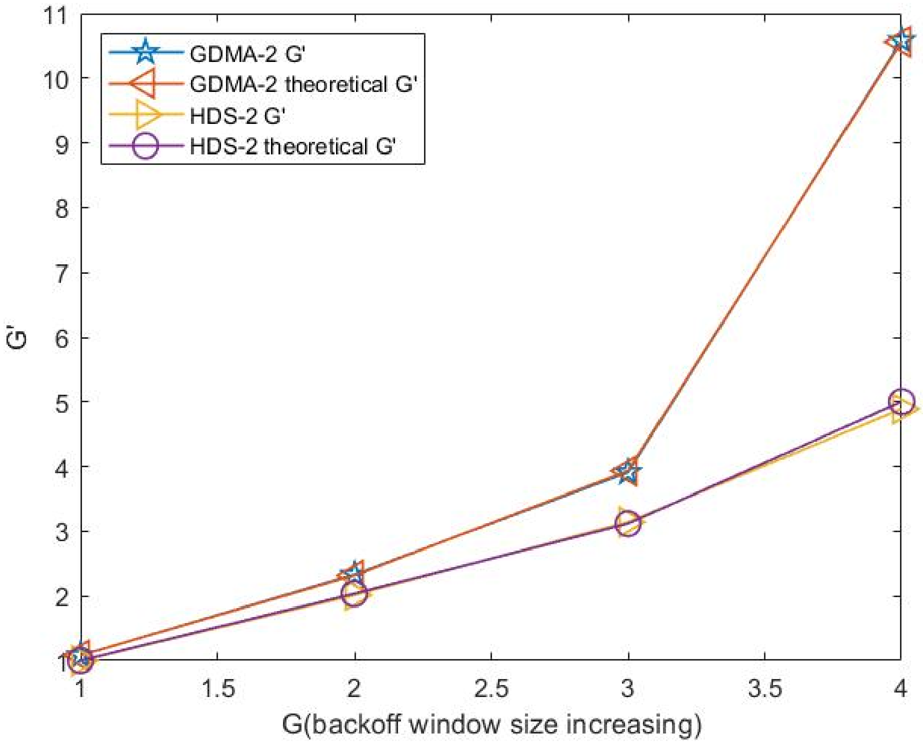

Figure 4.18: The relationship between G' and G with backoff window size $[0, 7]$, $[0, 15]$, and $[0, 31]$. SNR = 10 dB.

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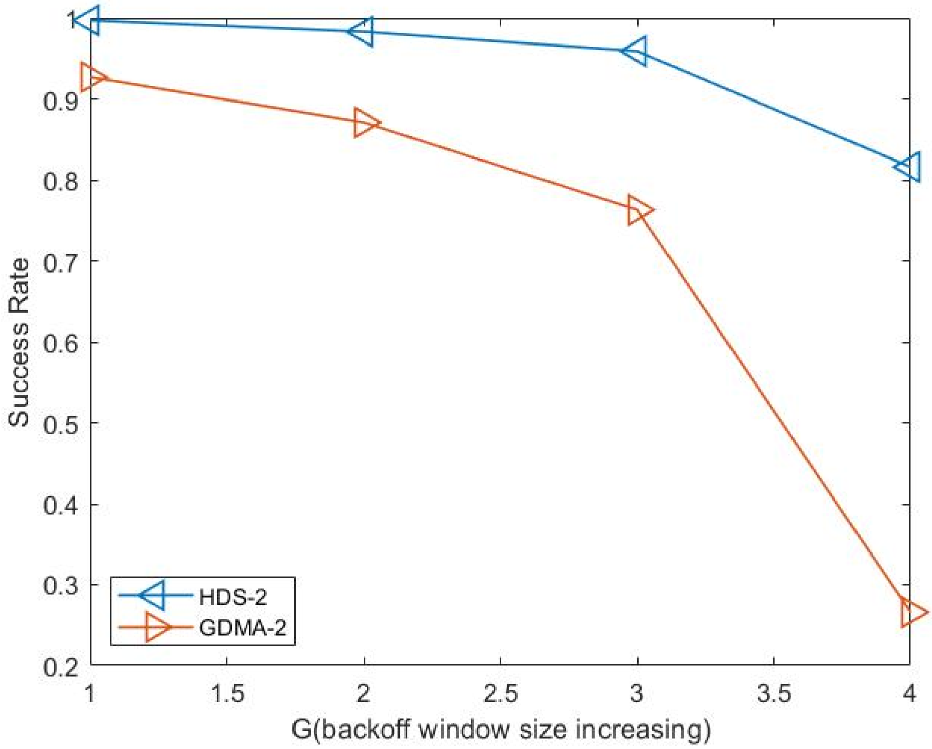

Figure 4.19: The success rate with backoff window size $[0, 7]$, $[0, 15]$, and $[0, 31]$. SNR = 10 dB.

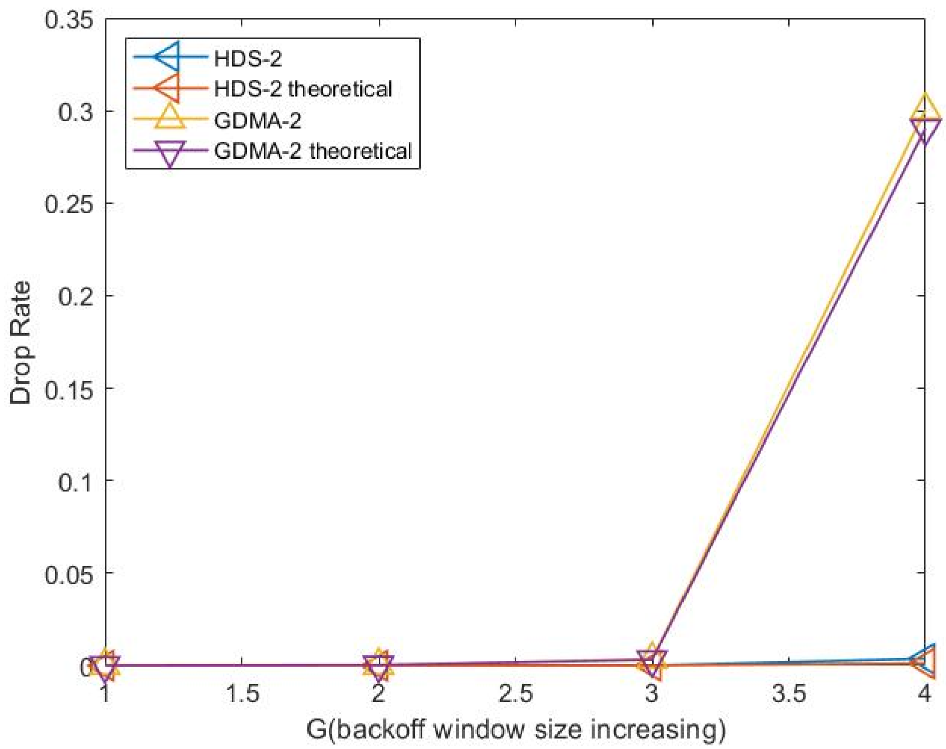

Figure 4.20: The drop rate with backoff window size $[0, 7]$, $[0, 15]$, and $[0, 31]$. SNR = 10 dB.

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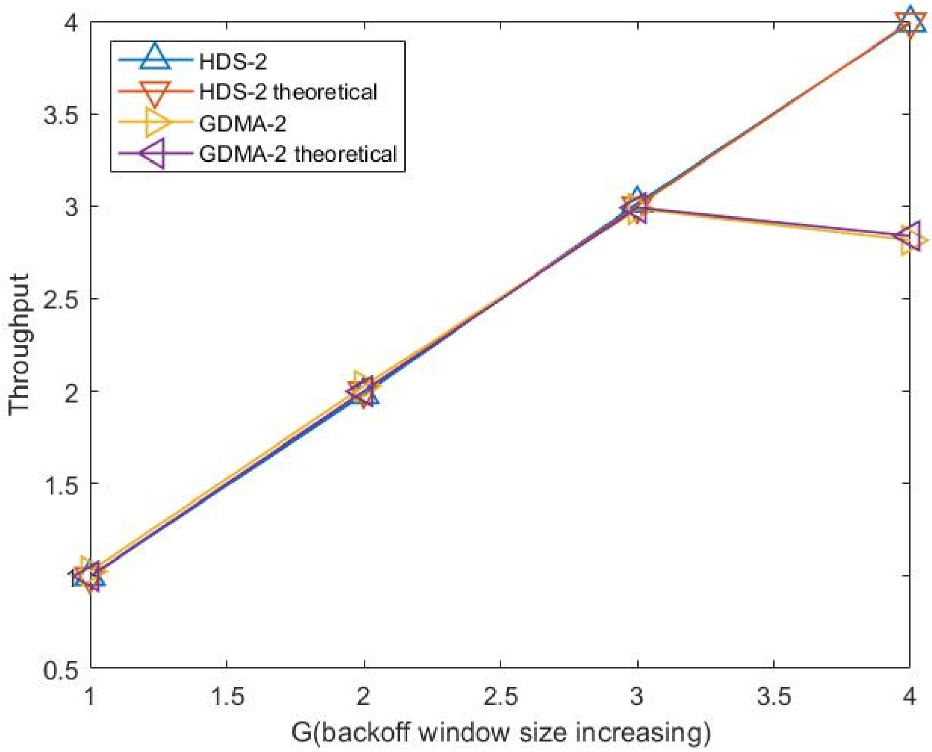

Figure 4.21: The throughput with backoff window size $[0, 7]$, $[0, 15]$, and $[0, 31]$. SNR = 10 dB.

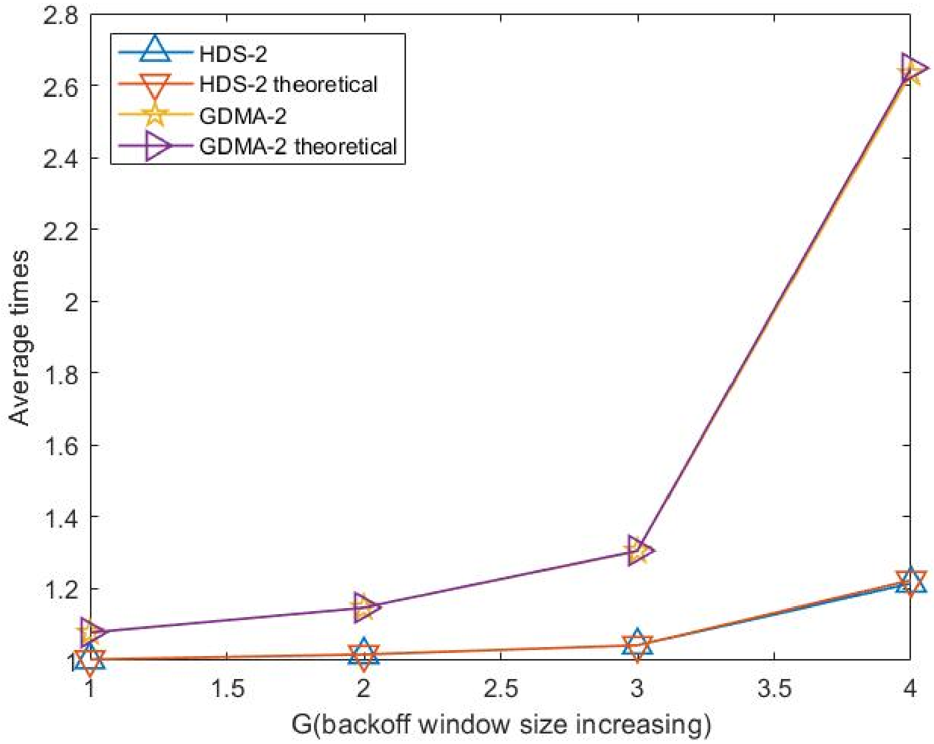

Figure 4.22: Average transmission attempts with backoff window size $[0, 7]$, $[0, 15]$, and $[0, 31]$. SNR = 10 dB.

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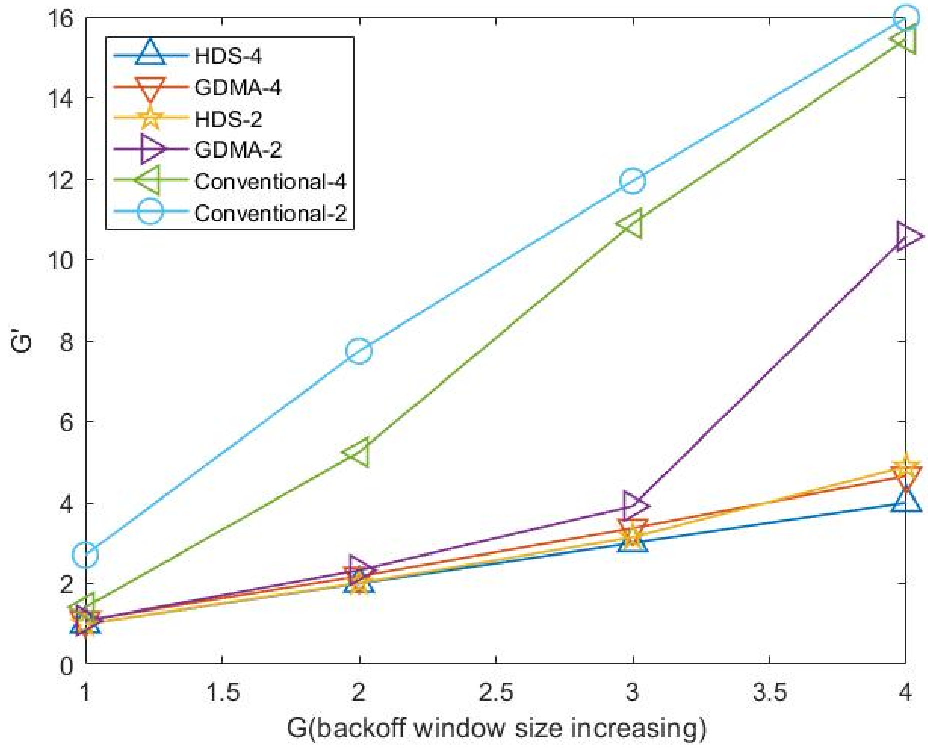

Figure 4.23: The relationship between G' and G with backoff window size $[0, 7]$, $[0, 15]$, and $[0, 31]$. SNR = 10 dB.

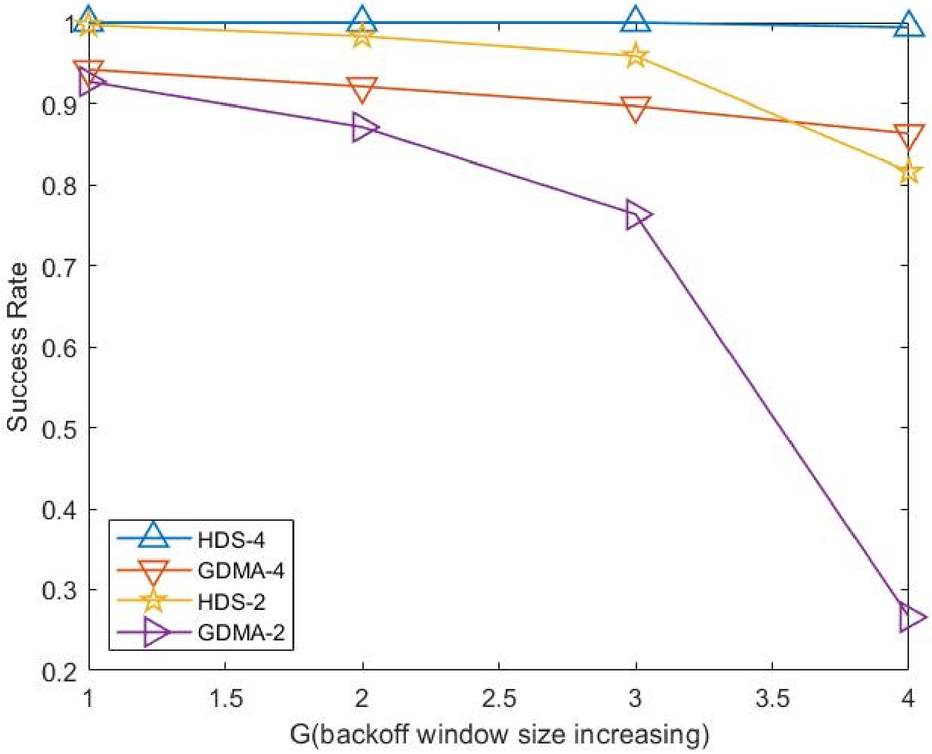

Figure 4.24: The success rate with backoff window size $[0, 7]$, $[0, 15]$, and $[0, 31]$. SNR = 10 dB.

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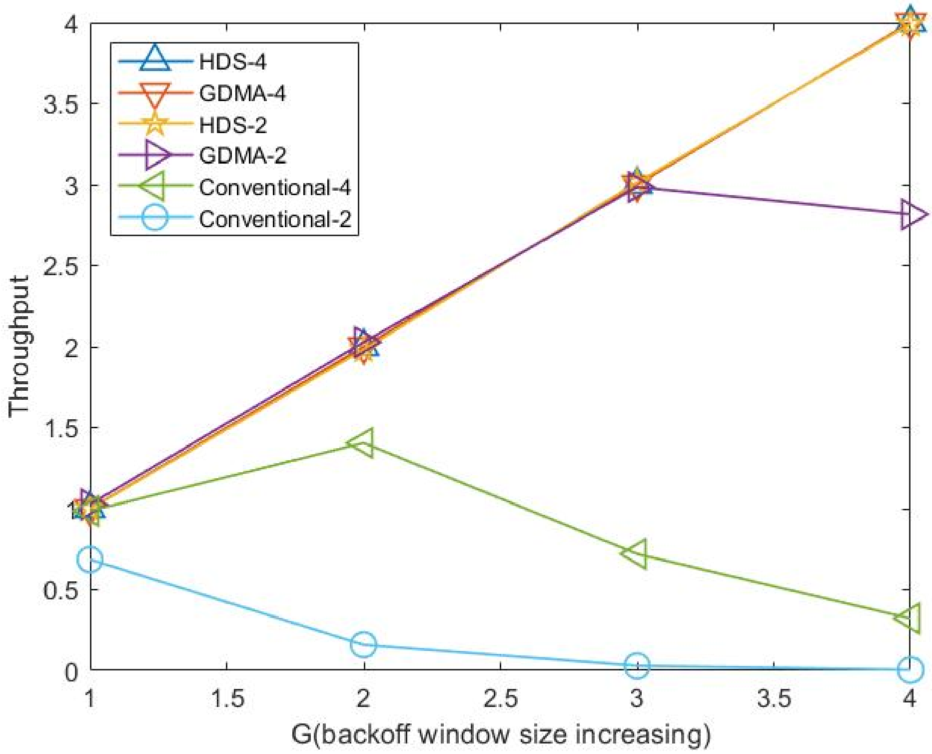

Figure 4.25: The throughput with backoff window size $[0, 7]$, $[0, 15]$, and $[0, 31]$. SNR = 10 dB.

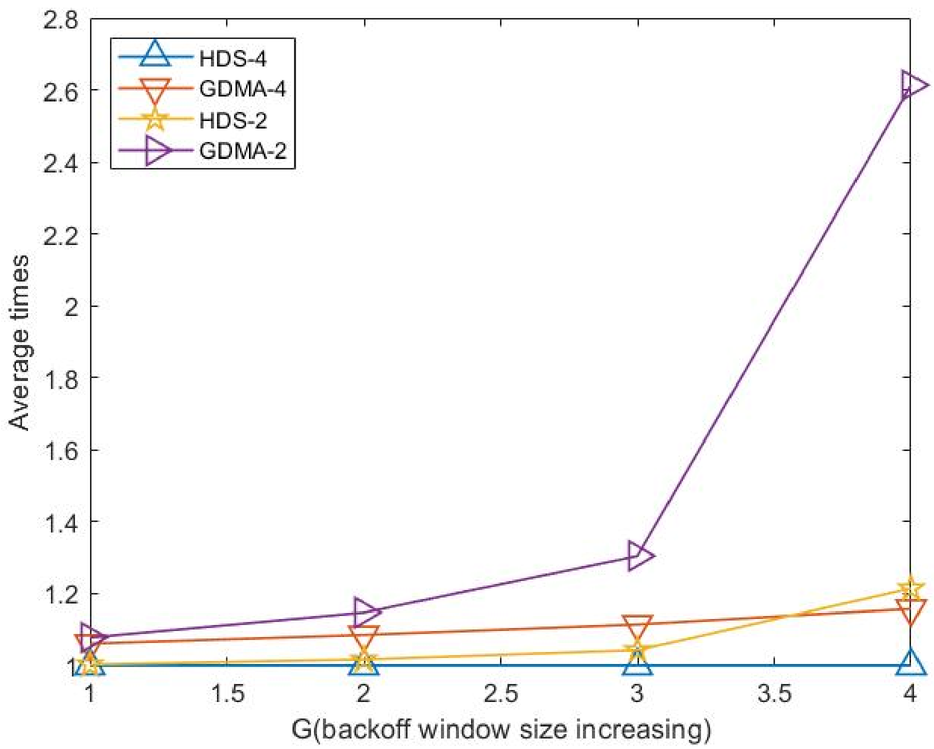

Figure 4.26: Average transmission attempts with backoff window size $[0, 7]$, $[0, 15]$, and $[0, 31]$. SNR = 10 dB.

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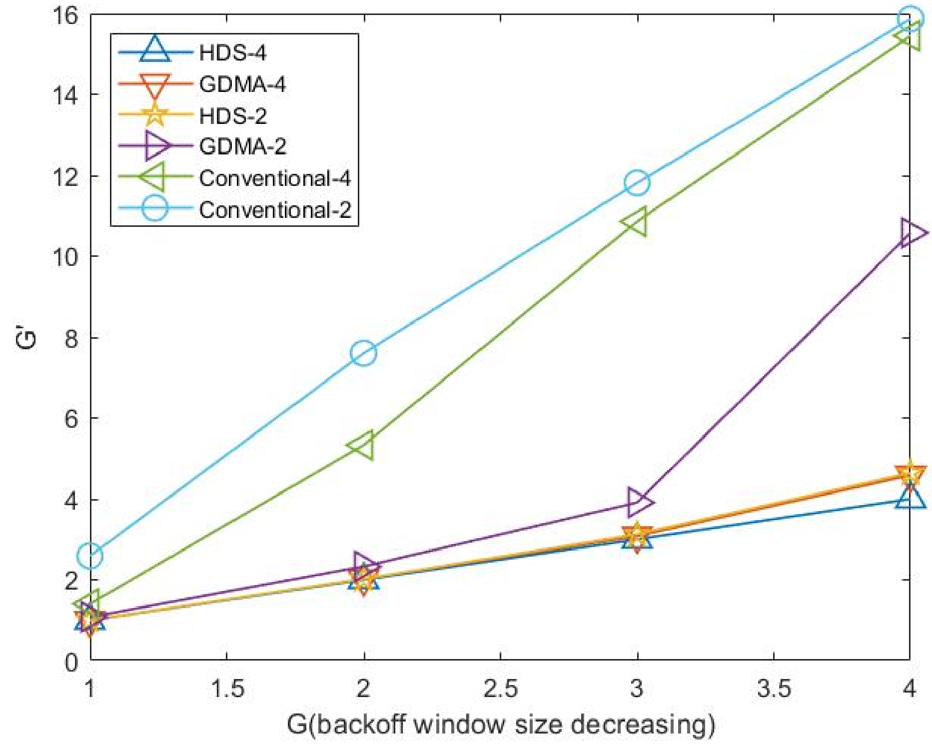

Figure 4.27: The relationship between G' and G with backoff window size $[0, 31]$, $[0, 15]$, and $[0, 7]$. SNR = 10 dB.

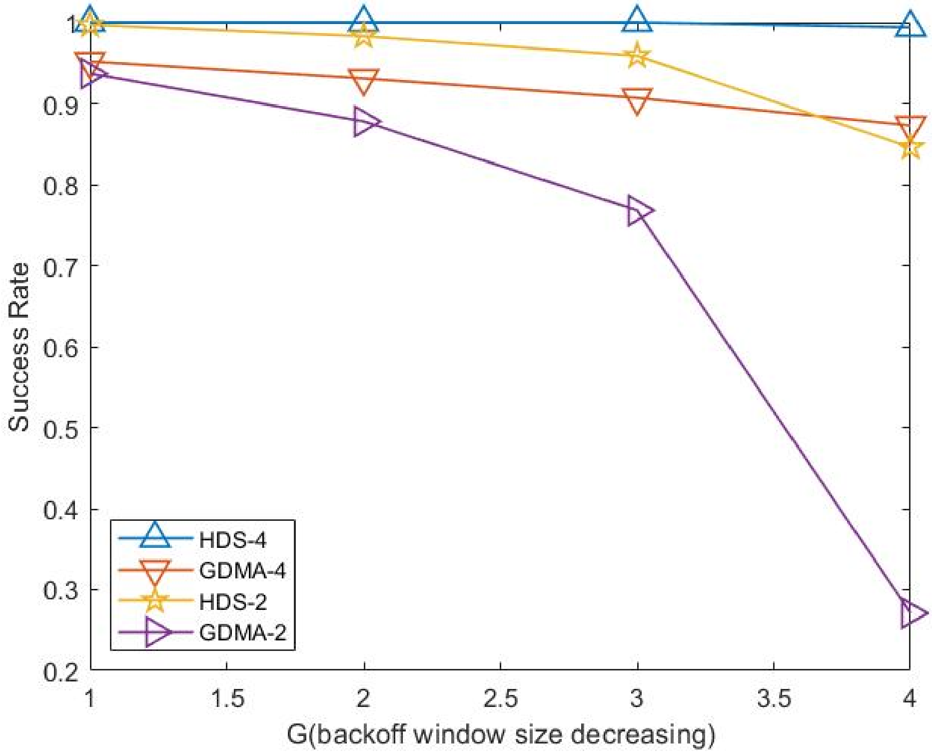

Figure 4.28: The success rate with backoff window size $[0, 31]$, $[0, 15]$, and $[0, 7]$. SNR = 10 dB.

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4.6 Slotted ALOHA with Finite Buffer

In previous sections, slotted ALOHA systems are all examined without taking into account packet queuing at user stations. In [21], it analyzes CDMA slotted ALOHA systems with finite buffer. We revise a bit and show our system model in Fig. 4.29.

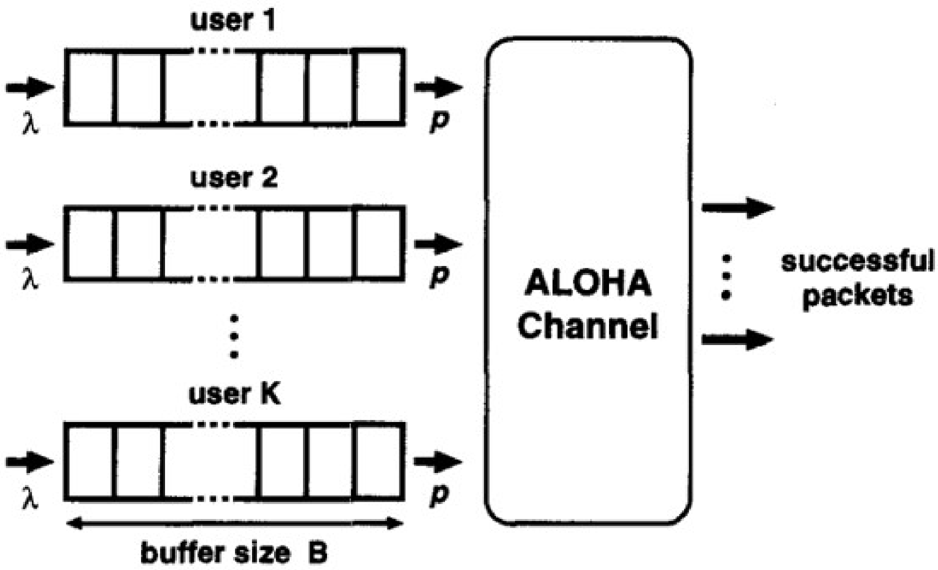

Figure 4.29: The ALOHA systems with finite buffer assumptions.

Previously, we considered a slotted ALOHA system operating under the assumption that the arrival process, as viewed from the hub station, follows a Poisson distribution. This model helped us understand the stochastic nature of packet arrivals and their implications on system performance. However, to delve deeper into the dynamics at the user level, we now shift our focus. Let's consider a scenario where each user station independently has a birth probability denoted by λ. This perspective allows us to explore the probabilistic behavior of individual stations.

First, examine a system with K user stations, where each station has a buffer capacity constrained to B packets. Each user station generates a packet in every time slot with a

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probability of λ. If a packet arrives at a station with a full buffer, the packet is discarded. Figure 4.30 illustrates the rejection probability as a function of the offered load. It is evident that increasing the buffer size leads to a reduction in rejection probability for lower offered load values.

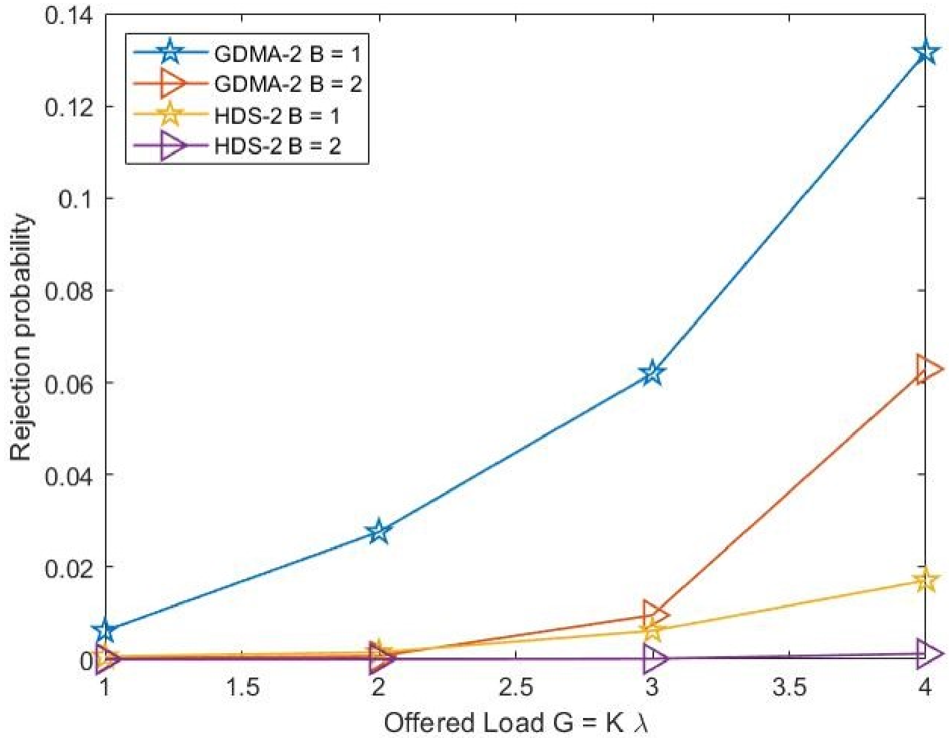

Figure 4.30: Rejection probability under different load.

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Chapter 5 Conclusions and Future Works

In Chapter 2 , we reviewed the principles of gain-division multiple access (GDMA), a method that differentiates user messages based on their unique channel gains. Additionally, we implemented GDMA using various modulation levels over a Rayleigh fading channel and analyzed the resulting error performance.

In Chapter 3, we reviewed low-density signature code division multiple access (LDS-CDMA) and high-density signature code division multiple access (HDS-CDMA), comparing the error performance of various systems decoded by different algorithms over a Rayleigh fading channel. Additionally, we applied error-correcting codes and OFDM to GDMA and HDS-CDMA and then denote OFDM-GDMA and OFDM-HDS-CDMA to enhance system reliability. We further expanded the resource number and simulated different usage of the resources, which we denote GS.

In Chapter 4, we reviewed fundamental concepts of the random access channel. The idea of combining OFDM-GDMA and OFDM-HDS-CDMA into ALOHA system is reviewed and denote OFDM-HDS-Q-ALOHA and OFDM-GDMA-Q-ALOHA in this thesis, where Q is the resource number. We utilized the idea and expanded the work under larger offered load G and larger resource number Q. We simulated and calculated theo-

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retical values of these systems and showed that OFDM-HDS-Q-ALOHA is suitable for most conditions but with greater complexity than OFDM-GDMA-Q-ALOHA. We further consider the retransmission of our slotted ALOHA systems to reflect the practical access network. Last, we consider our slotted ALOHA systems with finite buffer.

In future researches, it is possible to explore deeper into OFDM-GDMA-Q-ALOHA and OFDM-HDS-Q-ALOHA. With larger Q, there are more possible usages of resources. One may compare those different systems under different SNR and G. For the retransmission scheme, one could research into it and may derive mathematical values or bounds under different schemes. One may also try different backoff window size and retransmission limit K to look for the optimal ones. We only research a bit of our systems with finite buffer. If interested, one can look deeper into them.

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